

CHEMISTRY OF ADVANCED ENERGY STORAGE DEVICES FOR E-MOBILITY(18G6E14)

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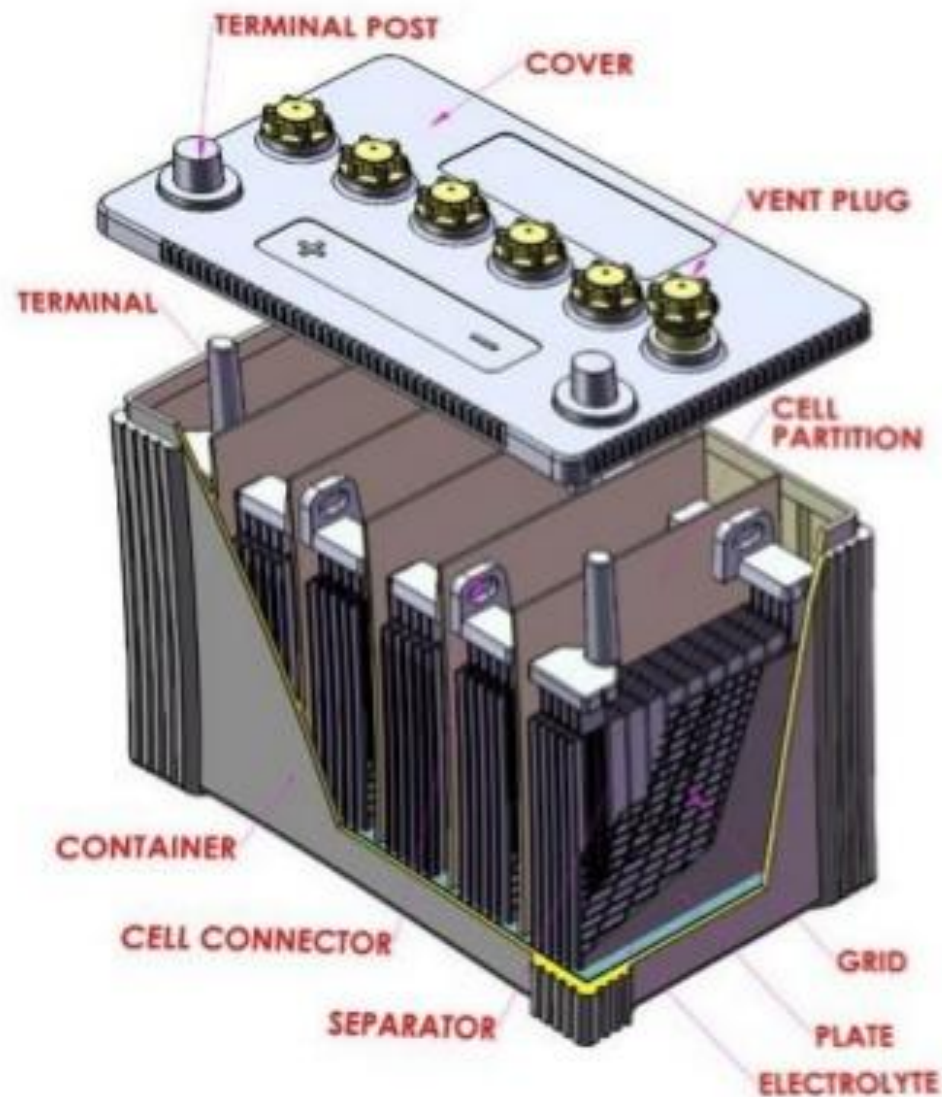
Unit II: Non-Lithium ion batteries



Limitations of Lithium ion batteries

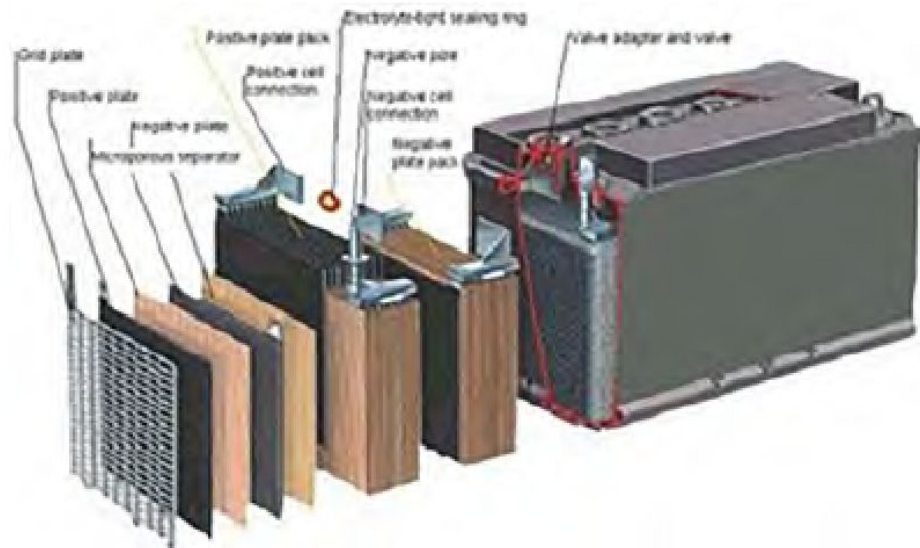
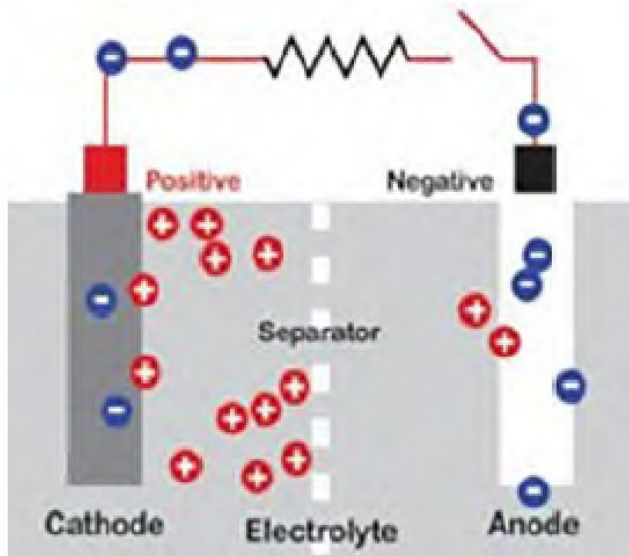
- Requires protection circuit to maintain voltage and current within safe limits.
- Subject to aging, even if not in use - storage in a cool place at 40% charge reduces the aging effect.
- Transportation restrictions - shipment of larger quantities may be subject to regulatory control. This restriction does not apply to personal carry-on batteries.
- Expensive to manufacture - about 40 percent higher in cost than nickel-cadmium.
- Not fully mature - metals and chemicals are changing on a continuing basis.

Lead–Acid Battery construction



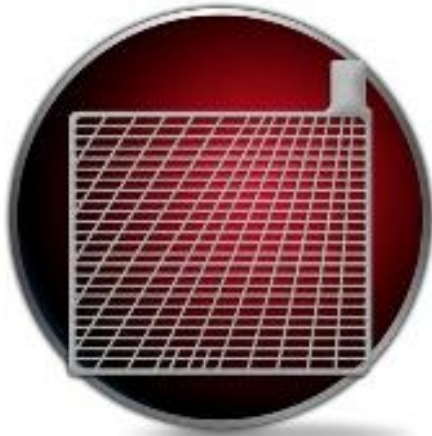
Lead–Acid (PbA) Battery

Figure E.1: Lead-Acid Battery Schematic and Physical Structure



Contd....

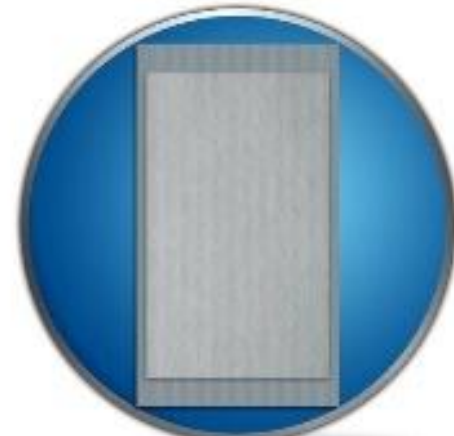
Grid – Enveloping – Separator & Fiberglass mat



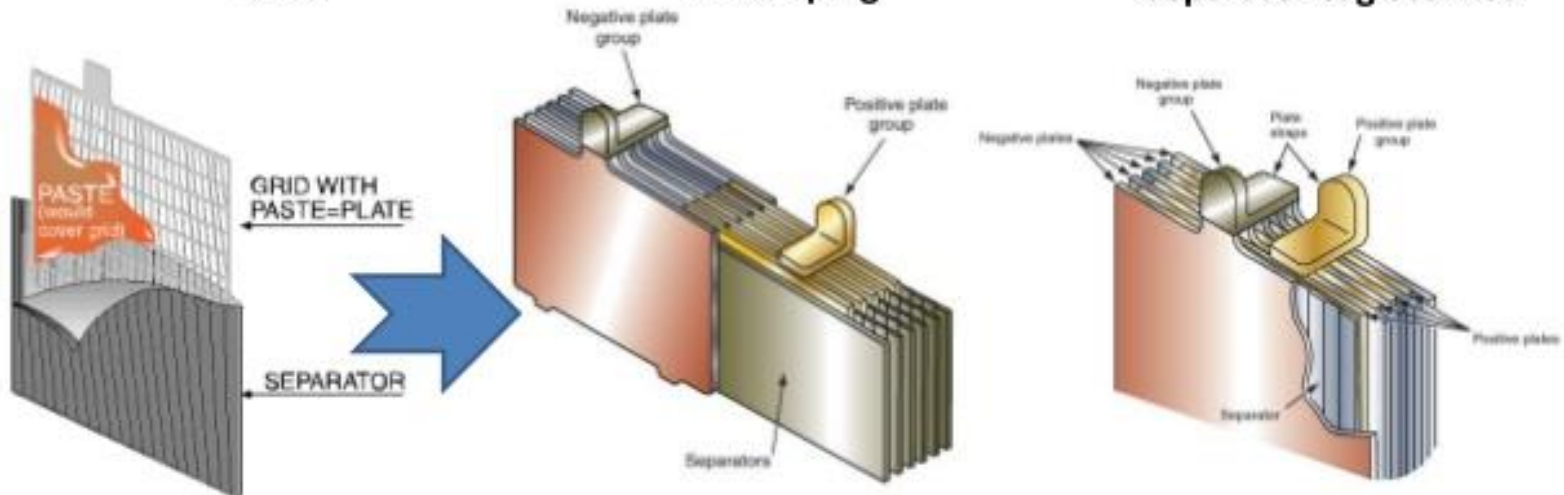
Grid



Enveloping

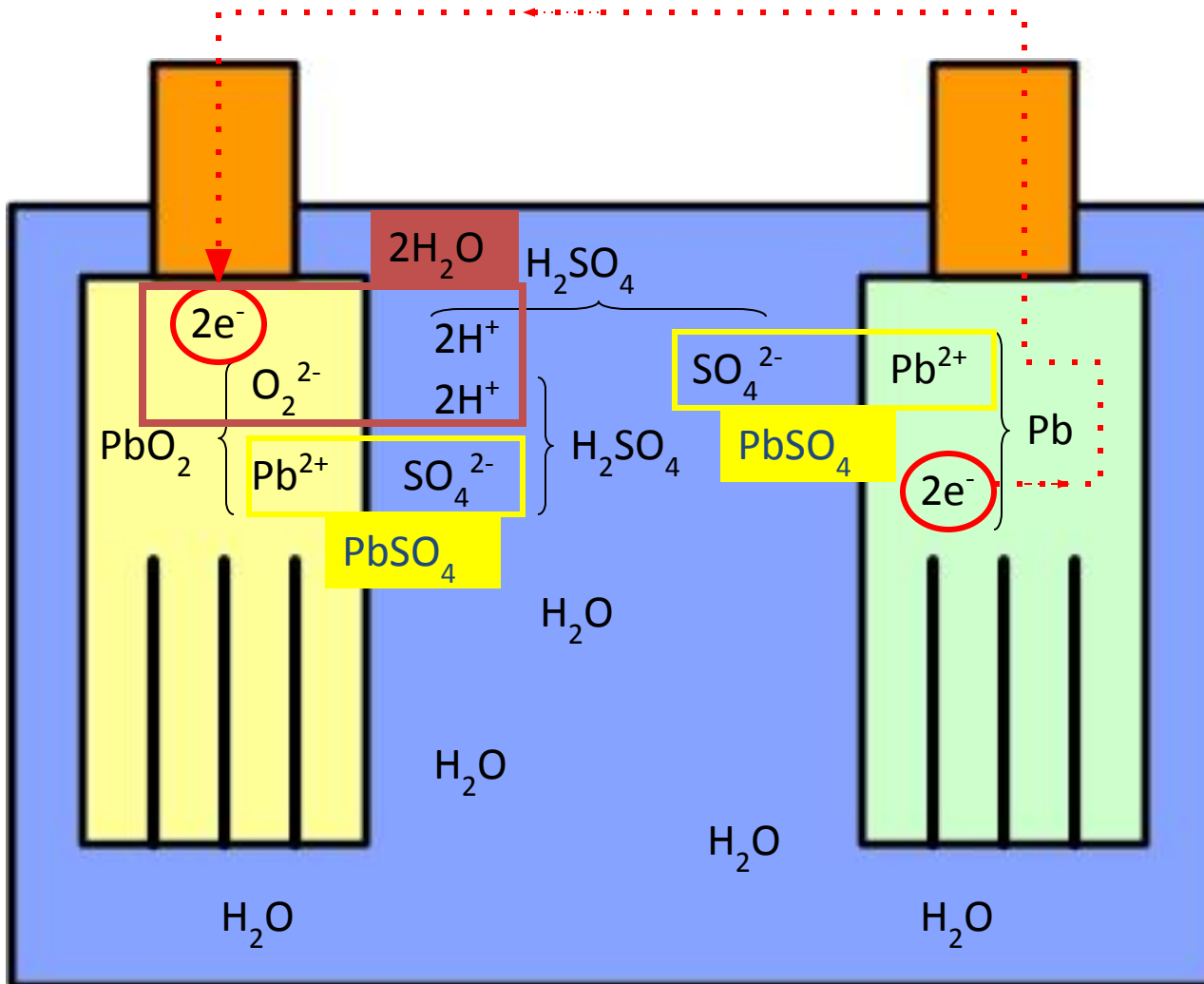


Separator & glass mat

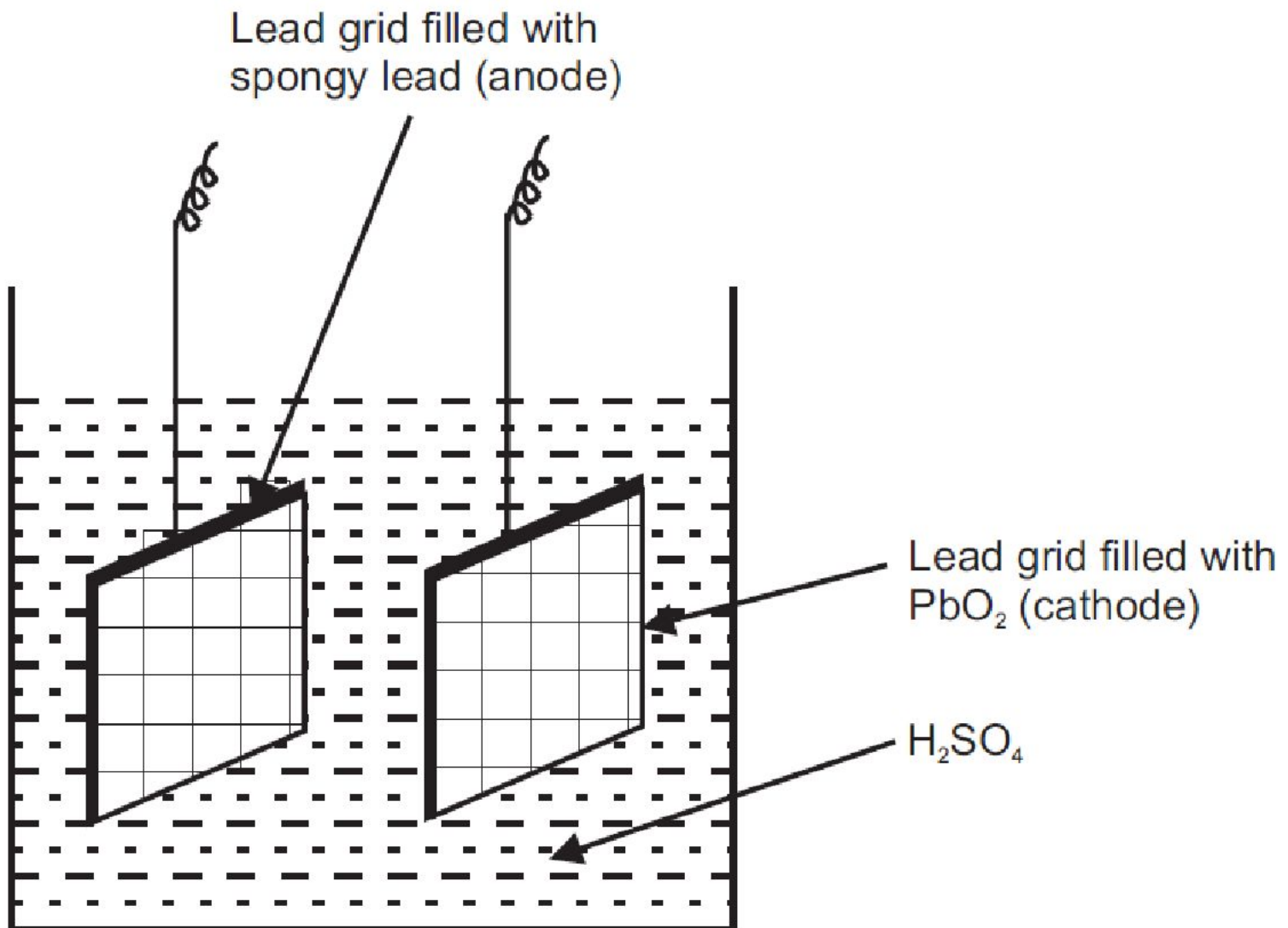


Lead-acid batteries

- Chemical reaction (discharge)



Lead–Acid (PbA) Battery





Lead-acid batteries Working

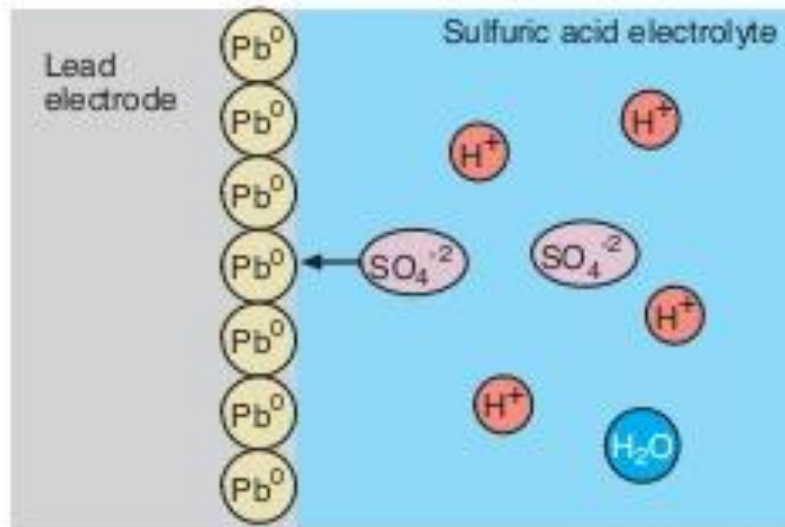
- As the battery discharges, sulfuric acid concentration decreases.
- At the same time, lead sulfate is deposited on the electrode plates.
- Charging follows the inverse process, but a small portion of the lead sulfate remains on the electrode plates.
- Every cycle, some more lead sulfate deposits build up on the electrode plates, reducing the reaction area and, hence, negatively affecting the battery performance.
- Electrode plates sulfatation is one of the primary effects that affects battery life.
- To avoid accelerating the sulfatation process, batteries need to be fully charged after every discharge and they must be kept charged at a float voltage higher than the nominal voltage. For lead acid batteries and depending their technology the float voltage is between 2.08 V/Cell and 2.27 V/cell. For the same reasons, lead-acid batteries should not be discharged below 1.75 V/cell

Conduction mechanism at anode

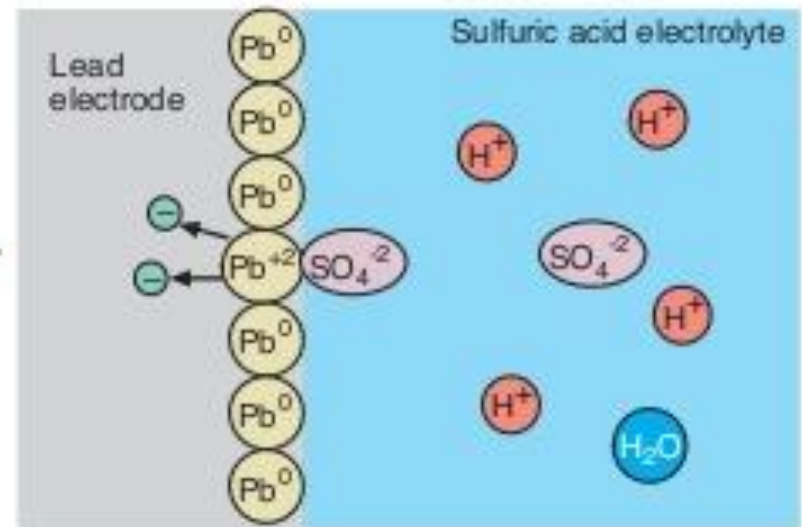
Oxidation-reduction (Redox) reaction transfers charge from ions in solution to conducting electrons in the electrode

At the surface of the lead (negative) electrode:

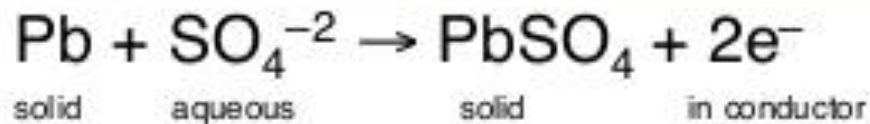
Charged sulfate ion approaches uncharged lead atom on surface of electrode



Lead atom becomes ionized and forms ionic bond with sulfate ion. Two electrons are released into lead electrode



Chemical reactions at lead electrode



This reaction releases net energy

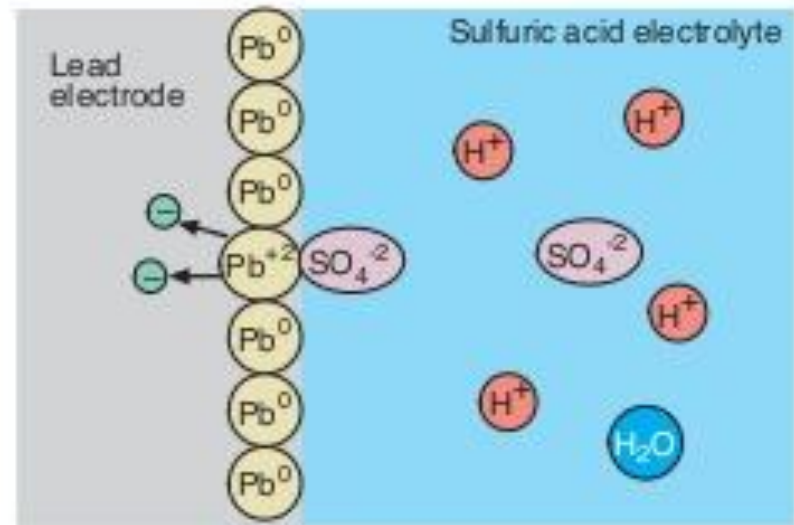
$$E^0 = 0.356 \text{ eV}$$

— the “Gibbs free energy”, under standard conditions ($T = 298^\circ\text{K}$, concentration = 1 molar)

Units: Energy = (charge)(voltage)

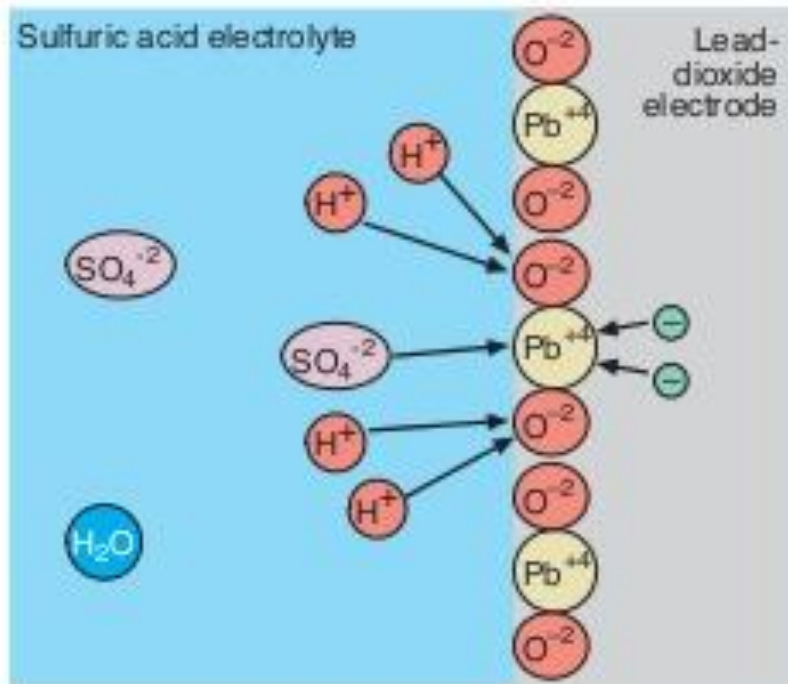
Energy in eV = (charge of electron)(1 V)

So the charge of the aqueous sulfate ion is transferred to two conducting electrons within the lead electrode, and energy is released.

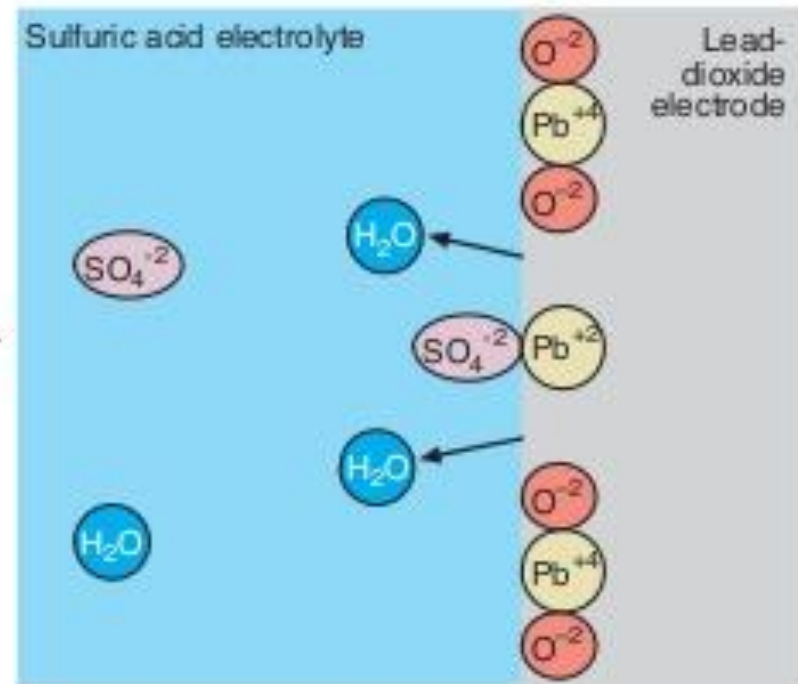


Conduction mechanism at cathode

Charged sulfate and hydrogen ions approach lead-dioxide molecule (net uncharged) on surface of electrode

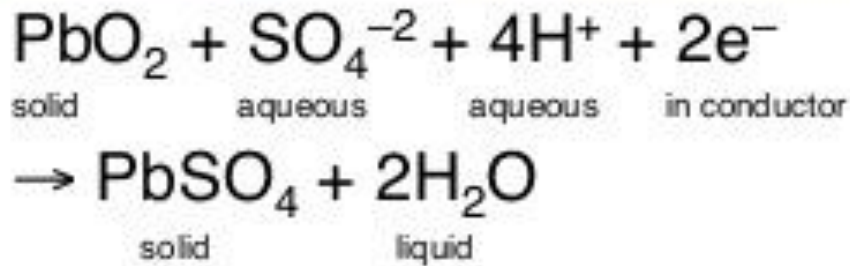


Lead atom changes ionization and forms ionic bond with sulfate ion. Two water molecules are released into solution



- Lead changes oxidation state from +4 to +2
- Two electrons are removed from conduction band in electrode

Chemical reaction at cathode

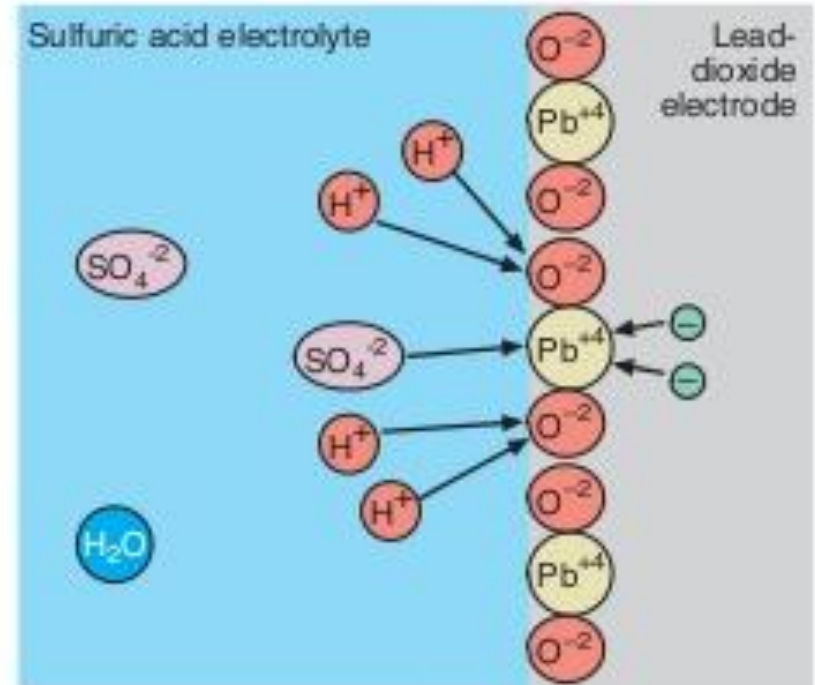


This reaction releases net energy

$$E^\circ = 1.685 \text{ eV}$$

Net charge of two electrons is transferred from the electrode into the electrolyte

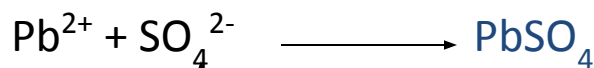
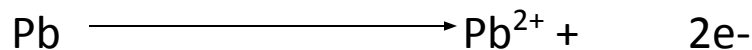
Both half reactions cause the electrodes to become coated with lead sulfate (a poor conductor) and reduce the concentration of the acid electrolyte



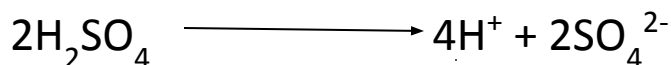
Lead-acid batteries Working

- Chemical reaction (discharge)

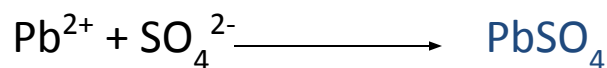
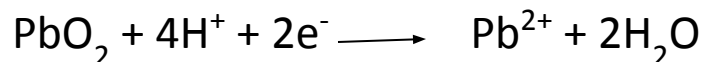
- Negative electrode



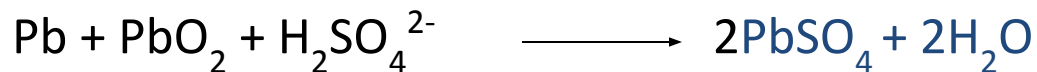
- Electrolyte



- Positive electrode



- Overall

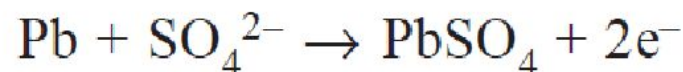
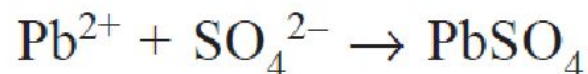


The nominal voltage produced by this reaction is about 2 V/cell. Cells are usually connected in series to achieve higher voltages, usually 6V, 12 V, 24 V and 48V.

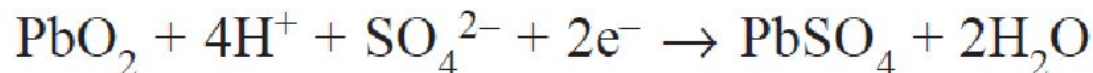
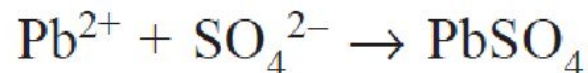
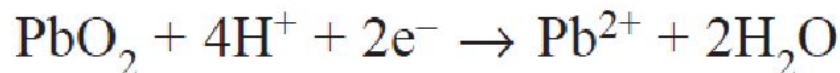


Lead-acid batteries Working

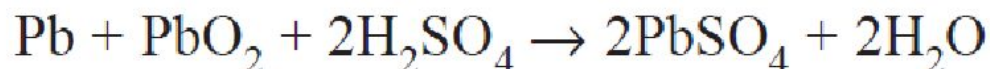
At the anode,



At the Cathode,

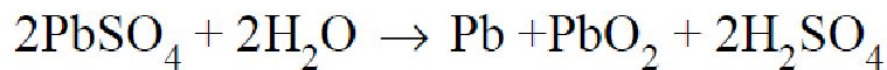
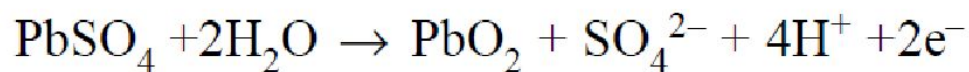
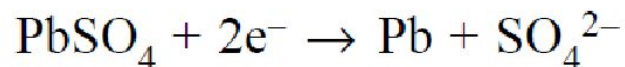


The net cell reaction is,



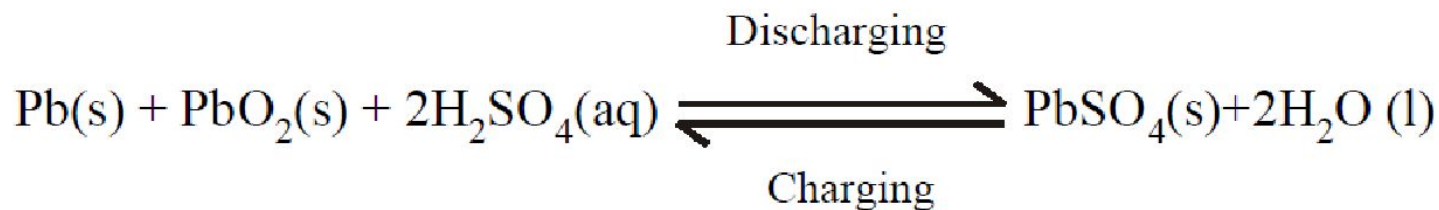


Lead–Acid (PbA) Battery



During charging, sulfuric acid is regenerated in the cell.

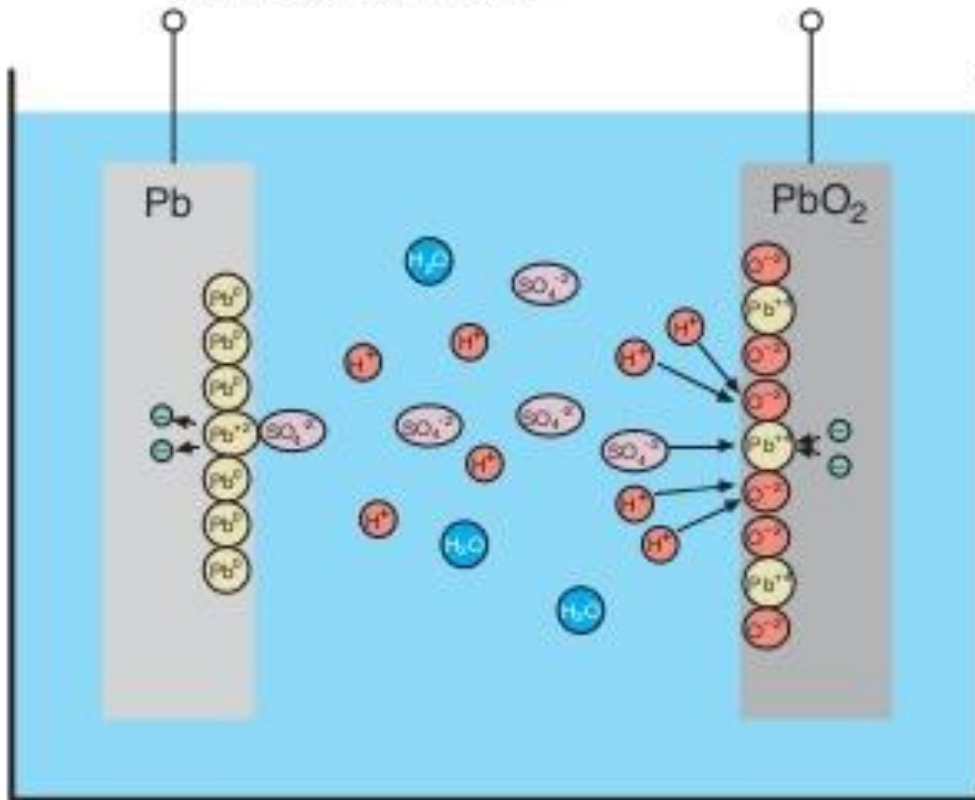
The net reaction during charging and discharging can be represented as follows:



How battery produces voltage

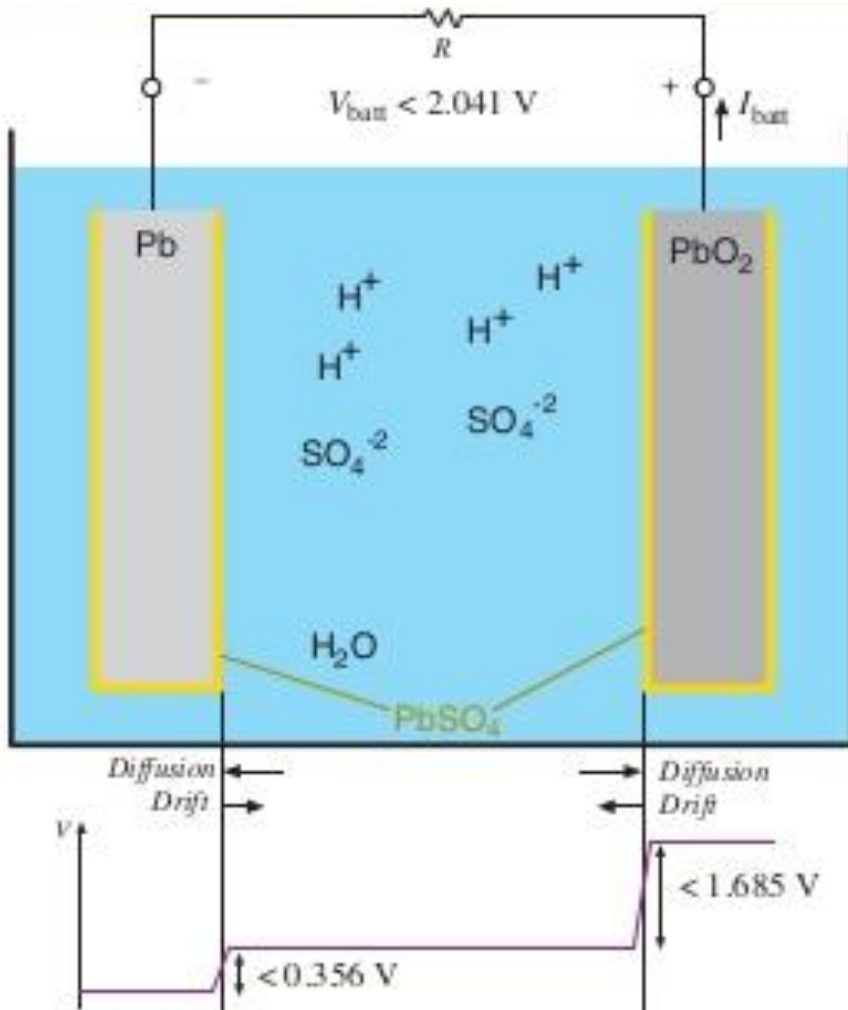
Consider the following experiment:

New electrodes are placed inside electrolyte, with no external electrical circuit connected



- The reactions start to occur
- They use up aqueous ions near electrodes
- Diffusion within electrolyte replenishes ions near electrodes
- Excess electrons are created in lead electrode, and electron deficit is created in lead-dioxide electrode
- Electric field is generated at electrode surfaces. This electric field opposes the flow of ions.

Discharging mechanism



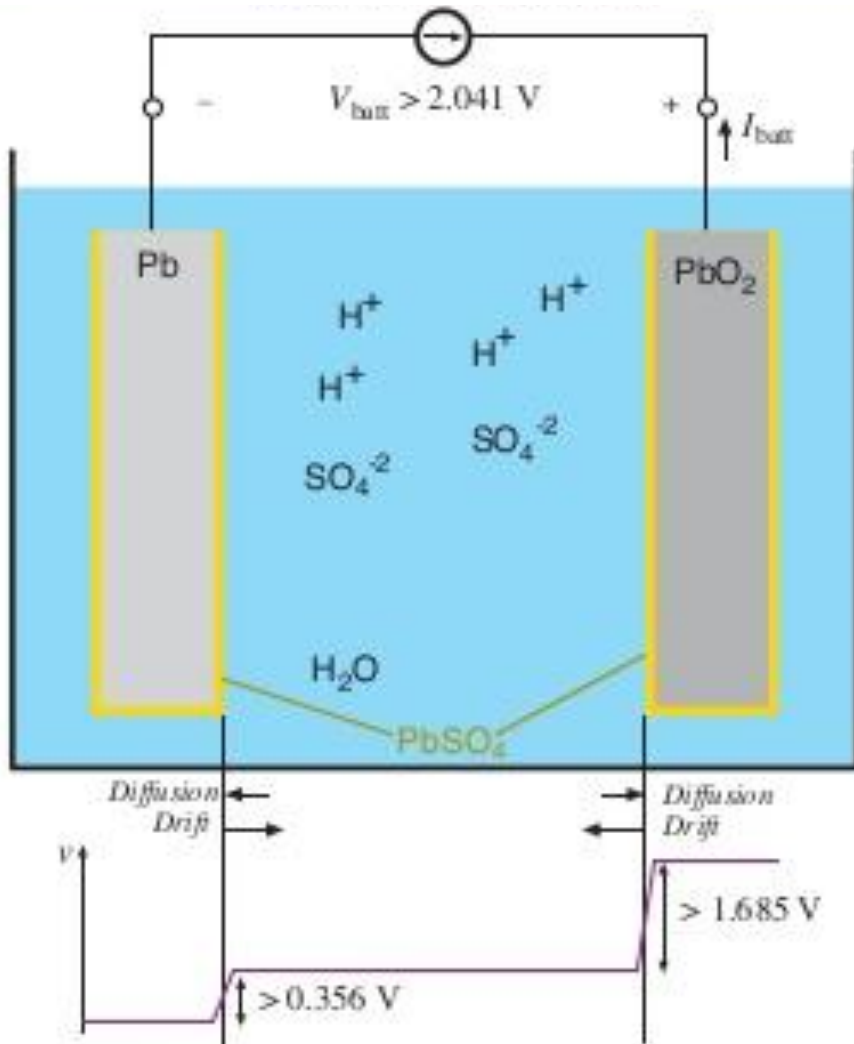
Connection of an electrical load allows electrons to flow from negative to positive terminals

This reduces the charge and the voltages at the electrodes

The chemical reactions are able to proceed, generating new electrons and generating the power that is converted to electrical form to drive the external electrical load

As the battery is discharged, the electrodes become coated with lead sulfate and the acid electrolyte becomes weaker

Charging mechanism



Connection of an electrical power source forces electrons to flow from positive to negative terminals

This increases the charge and the voltages at the electrodes

The chemical reactions are driven in the reverse direction, converting electrical energy into stored chemical energy

As the battery is charged, the lead sulfate coating on the electrodes is removed, and the acid electrolyte becomes stronger



Lead–Acid (PbA) Battery

Applications: Each electrode pair develops a potential of 2V. In order to obtain higher potential a number of electrode pairs are connected in series.

Lead storage batteries are extensively used in automobiles to start the engine. They are also used for electric supply in telephone exchangers, railway trains, hospitals, laboratories, etc., and for emergency power supplies.

Limitations:

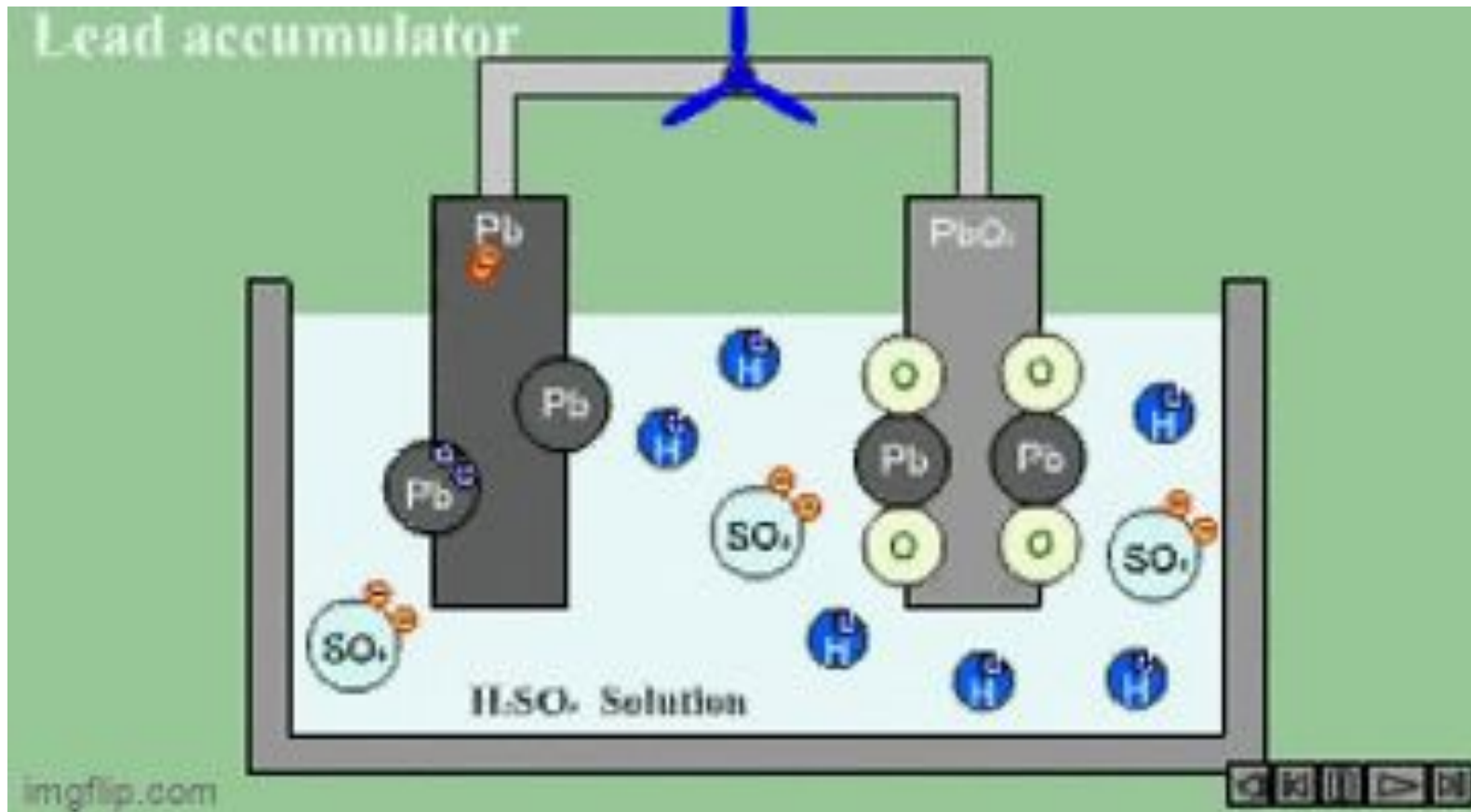
- The potential decreases with decrease in concentration of sulfuric acid.
- Excessive discharge and quick charging shortens the life of the battery.
- Cell potential and the effectiveness is reduced at low temperature.
- Excessive charging may damage the electrodes and may also lead to explosion.



Lead–Acid (PbA) Battery

Advantages	Disadvantages
Low-cost and simple manufacture	Low specific energy; poor weight-to-energy ratio
Low cost per watt-hour	Slow charging: Fully saturated charge takes 14–16 hours
High specific power, capable of high discharge currents	Need for storage in charged condition to prevent sulfation
Good performance at low and high temperatures	Limited cycle life; repeated deep-cycling reduces battery life
No block-wise or cell-wise BMS required	Watering requirement for flooded type
	Transportation restrictions for flooded type
	Adverse environmental impact

Lead–Acid (PbA) Battery: discharging

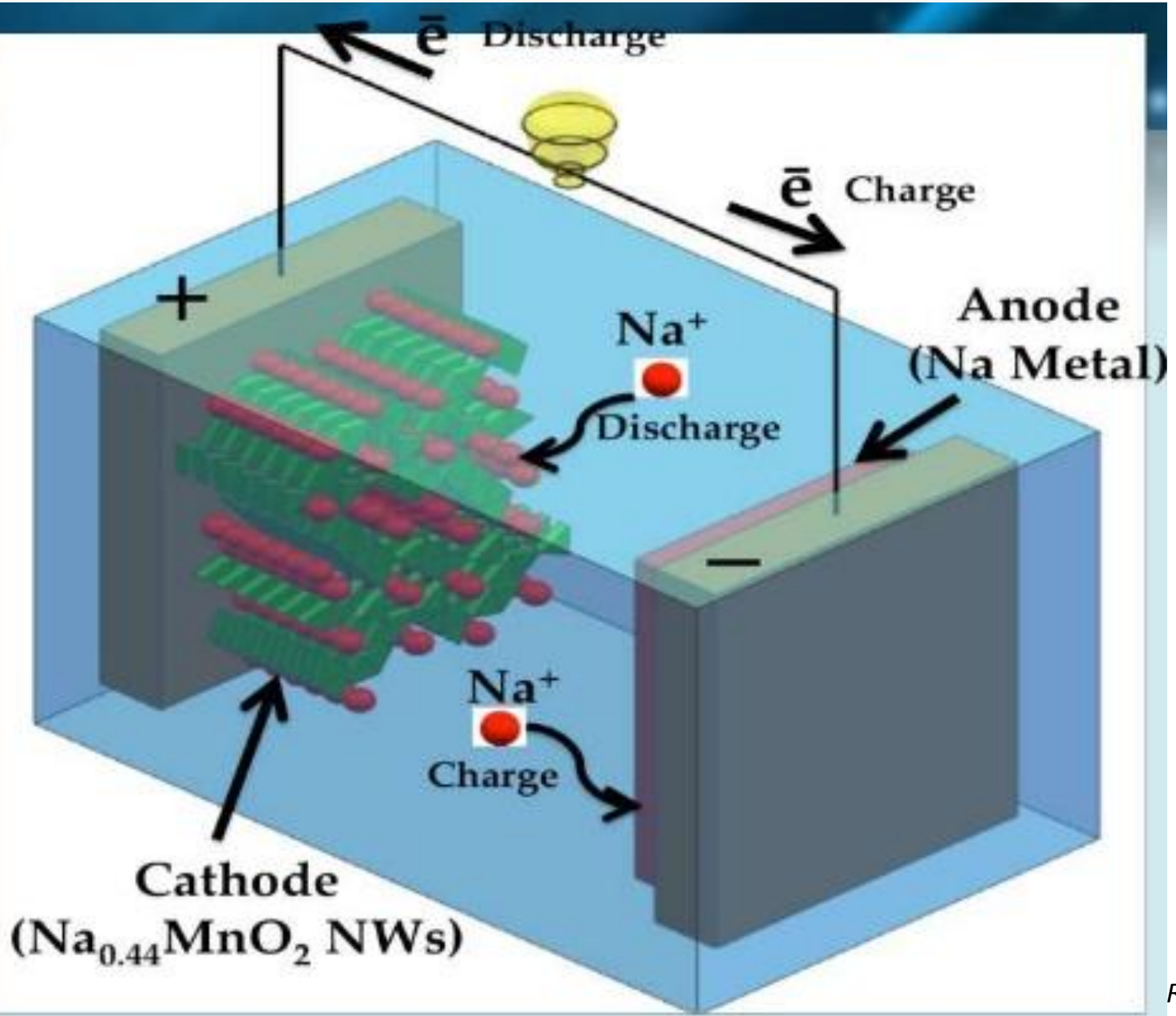




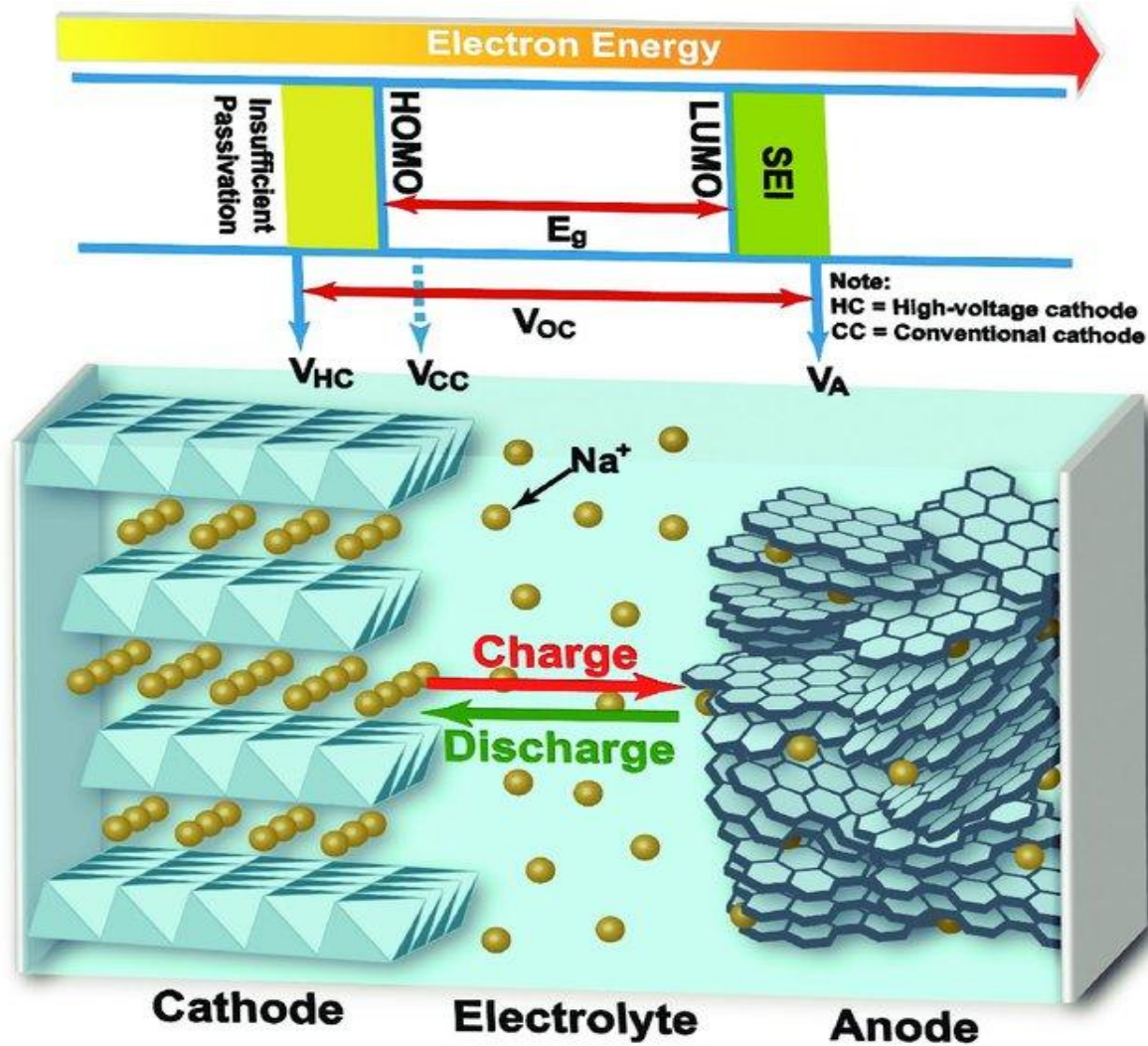
Sodium ion batteries Battery

- **Sodium-ion batteries** are a type of rechargeable battery that uses sodium ions as charge carriers.
- Sodium-ion battery is relatively young compared to other battery types.
- The battery-grade salts of sodium are cheap and abundant, much more than those of lithium.
- The first successful attempt of a sodium battery was undertaken in 1967 by Ford Motor Company (USA) in the sodium-sulfur battery.

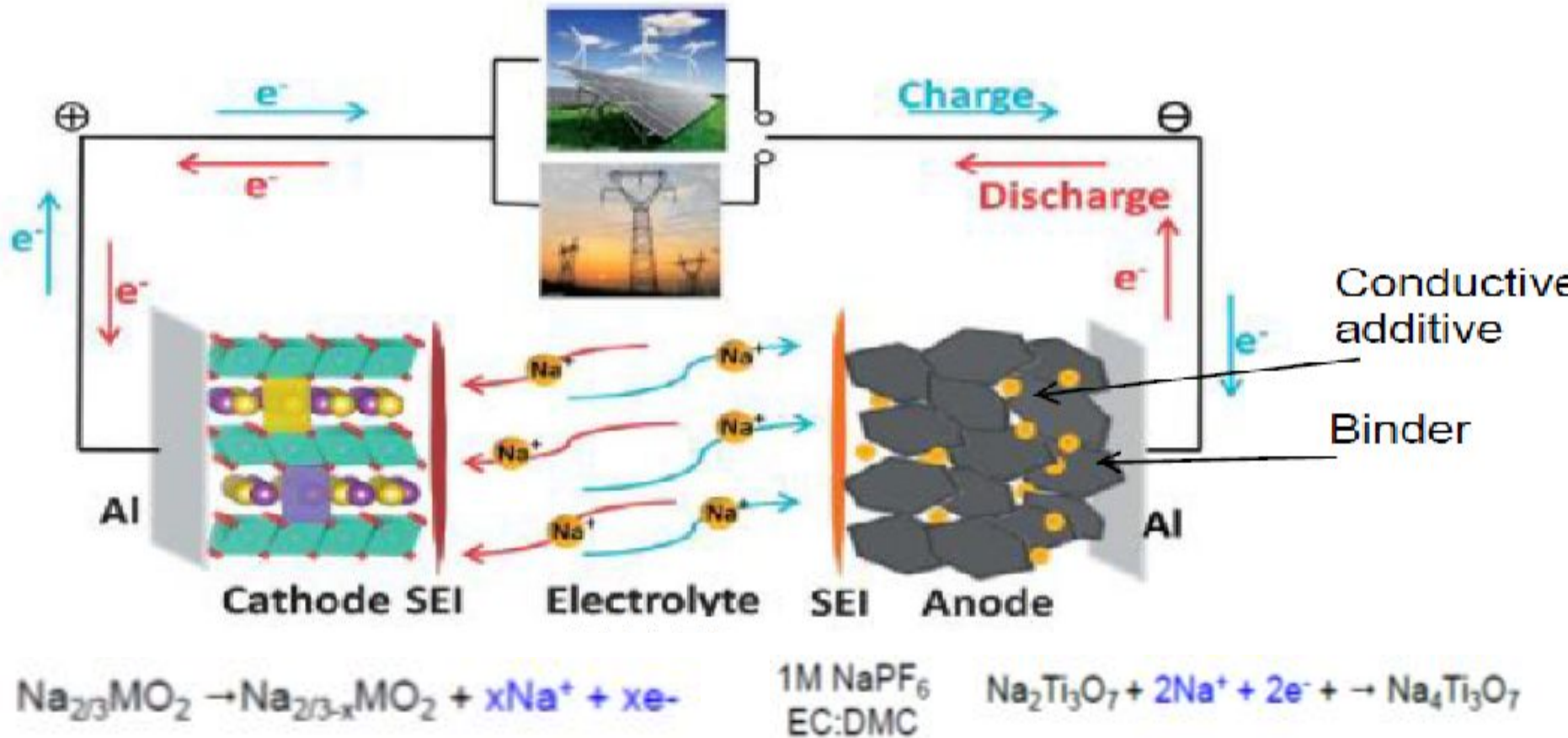
Sodium ion batteries



Sodium ion batteries



Sodium ion cells



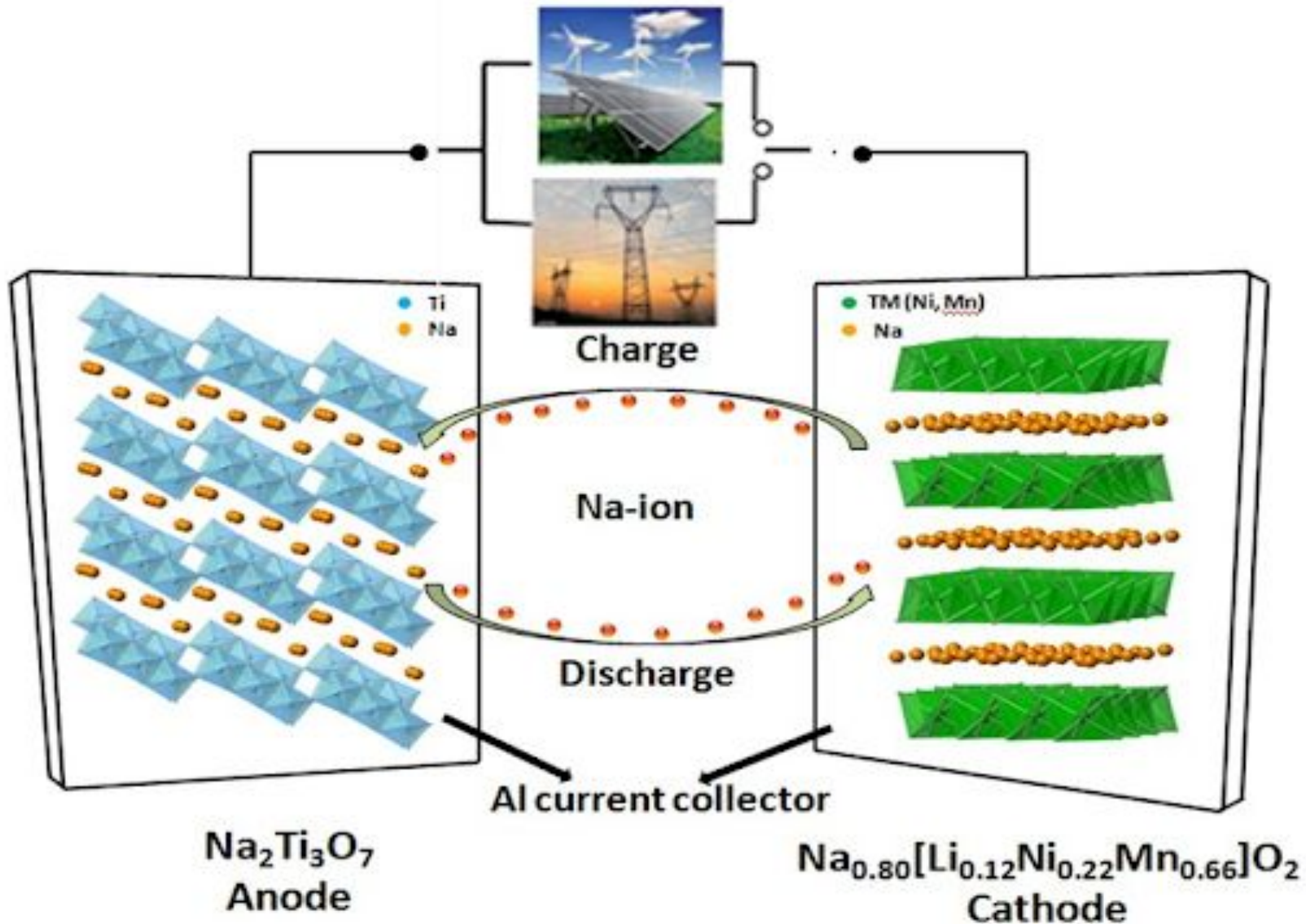
Ideal Battery

- ❖ High cell voltage and specific capacity
- ❖ High energy density
- ❖ Safety
- ❖ Low-cost
- ❖ Long cycle life.

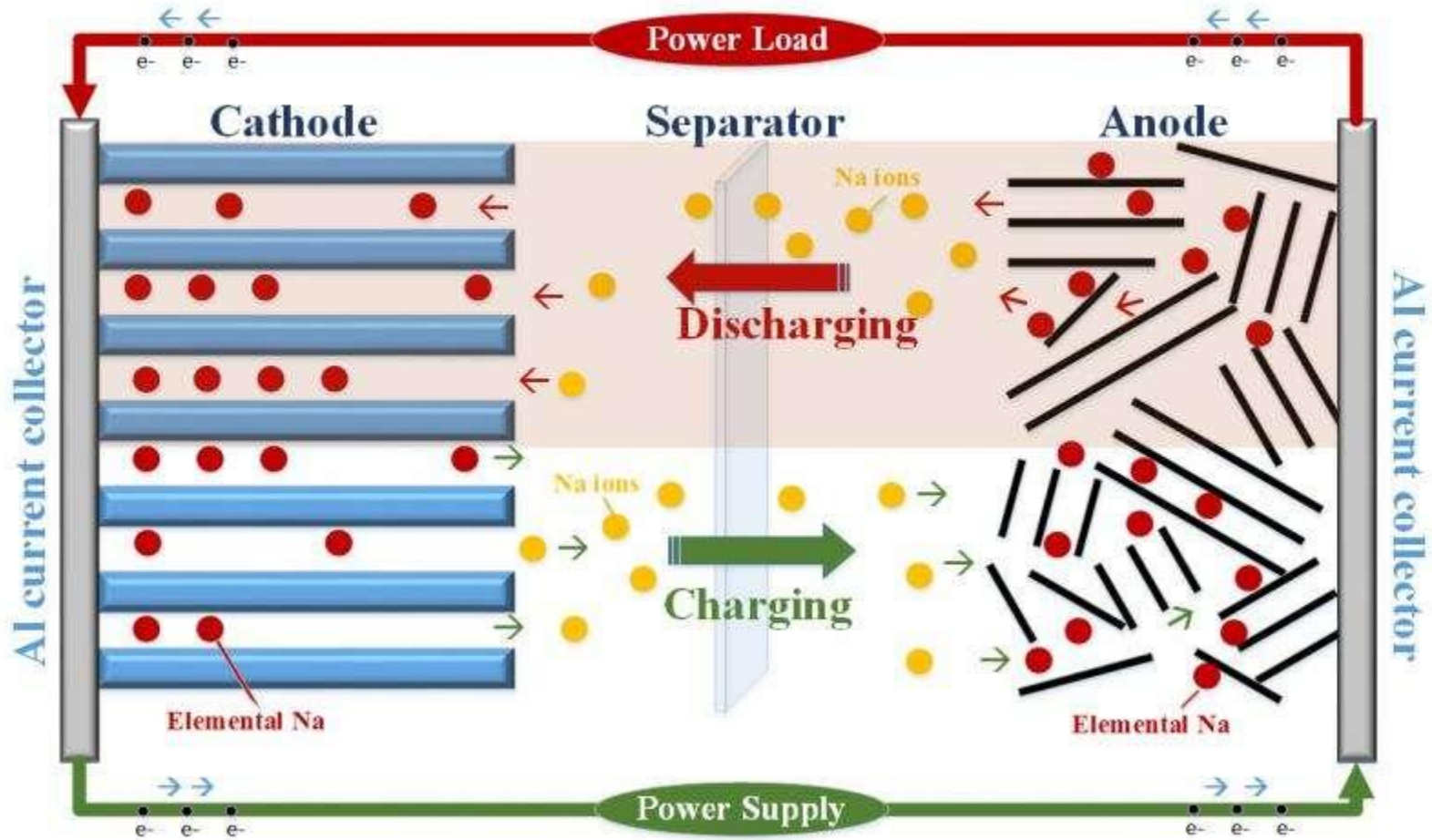
Modeling Calculations

- Energy Density : 210 Whkg⁻¹
- With:
- Cathode capacity of 200 mAhg⁻¹
- Anode : 500 mAhg⁻¹
- Average Cell Potential: 3.3V

Sodium ion cells



Sodium ion cells





Sodium ion battery components

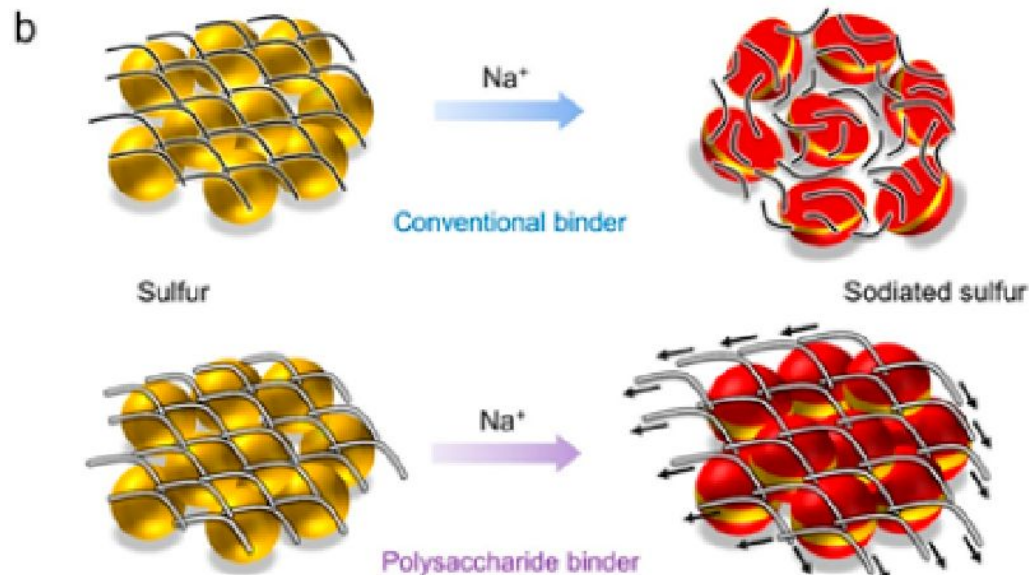
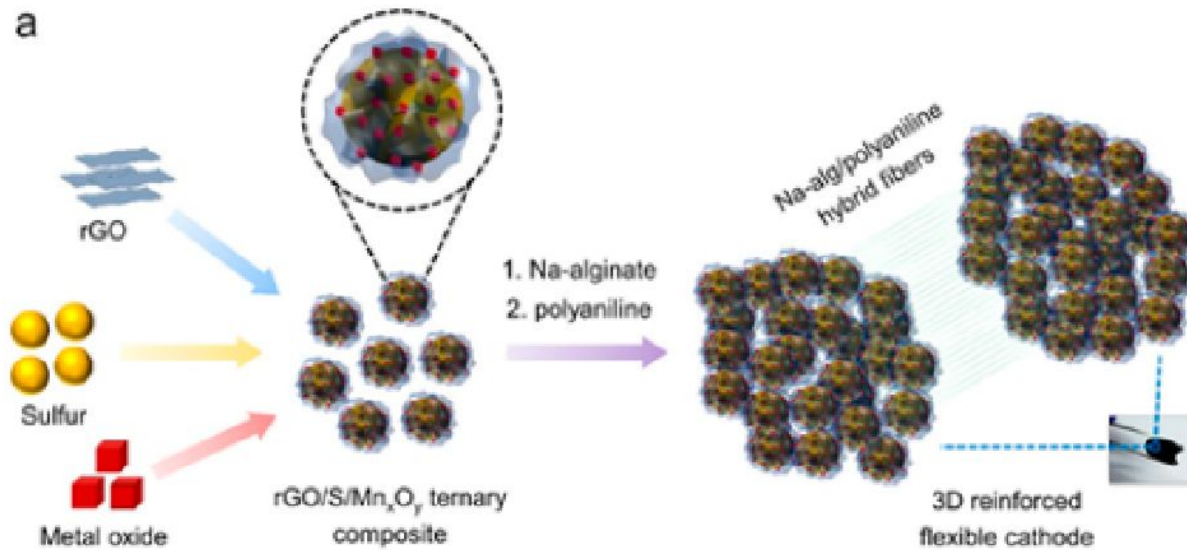
- They are many kinds of cathode materials for Na ion cells that have been examined, such as chalcogenides, fluorides, polyanion compounds and oxide compound. also chromium cathodes have been tested as cathodes.
- **Sodium fluorophosphates** is tested for new cathode materials.
- Different electrolyte are used for Na ion batteries such as: ethylene carbonate (EC), diethyl carbonate (DEC), propylene carbonate (PC), dimethyl carbonate (DMC), ethyl methyl carbonate (EMC), butylene carbonate (BC), and vinylene carbonate (VC) and salt combinations: LiPF_6 , and NaClO_4 .



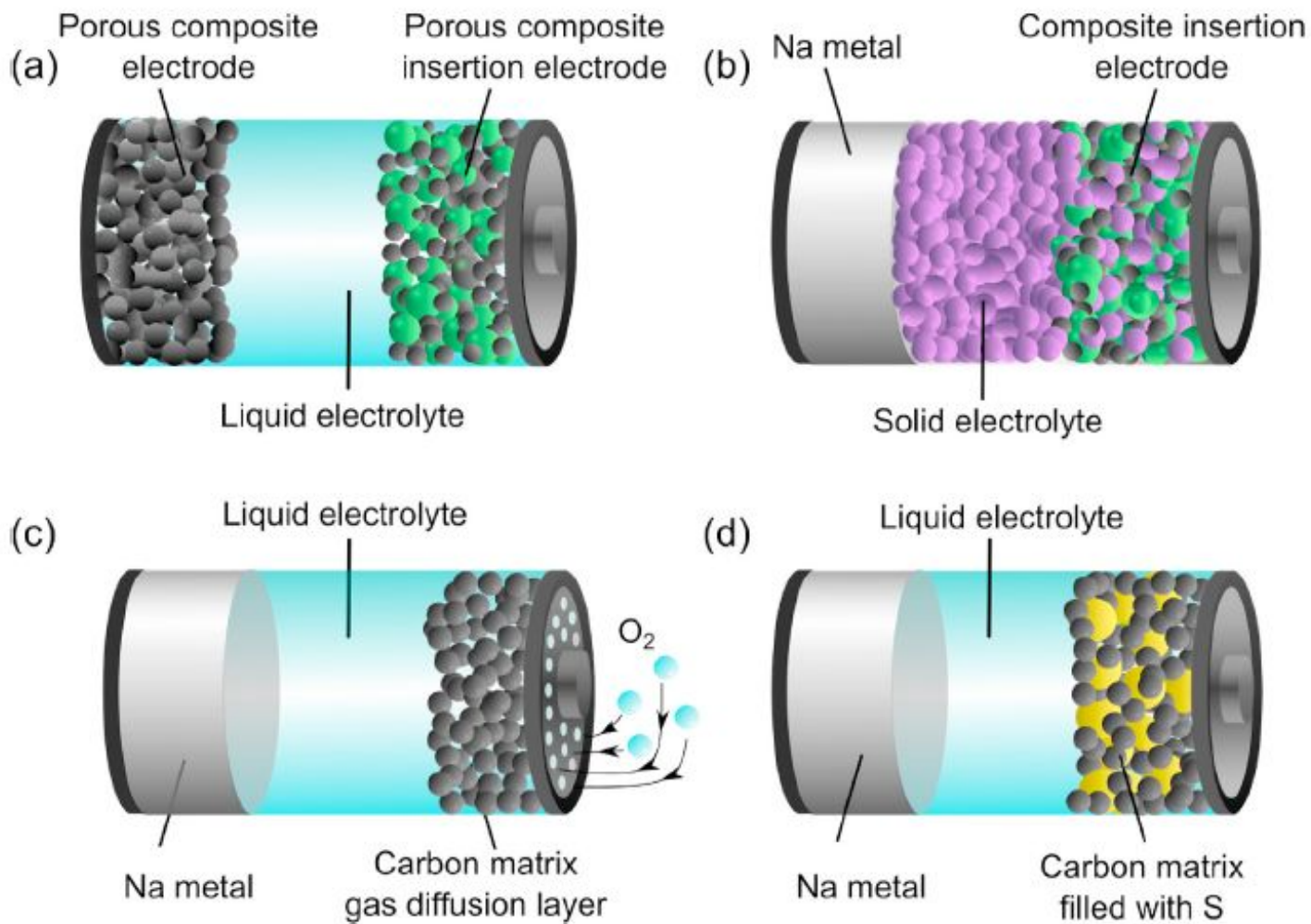
Contd...

- Porous carbon has been identified as one for anode material. In porous carbon, Na ions can be reversibly inserted into nanopores or absorbed on the surface of carbon, providing capacity of less than 200 mA h g^{-1} . In addition to carbon anodes, alloying different types of anode with additives such as **Antimony (Sb)**, **Tin (Sn)**, **Phosphorus (P)**, **Germanium (Ge)** and **Lead (Pb)** can also yield results.

Cathode materials



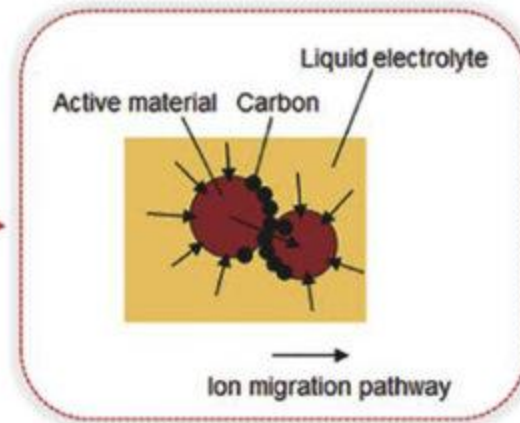
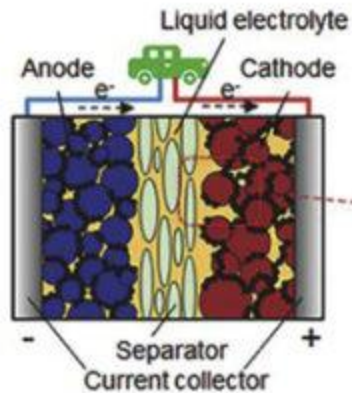
Sodium ion battery Types



Sodium ion battery Types

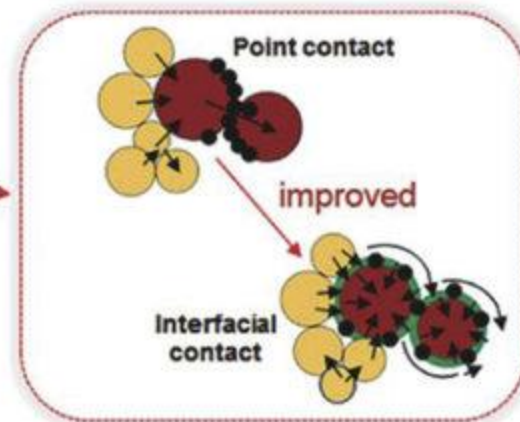
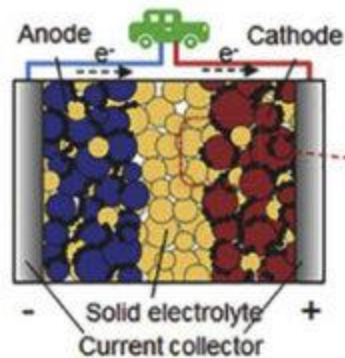
(a)

Conventional
Sodium-ion
Batteries



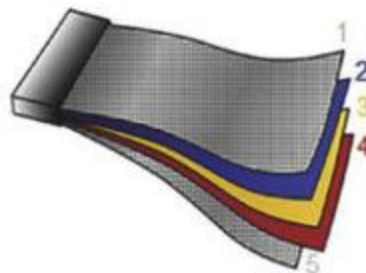
(b)

Solid-state
Sodium
Batteries



(c)

Flexible
Solid-state
Sodium
Batteries



- 1 Current collector (mesh)
- 2 Anode (Plastic)
- 3 Plastic electrolyte
- 4 Cathode (Plastic)
- 5 Current collector (mesh)

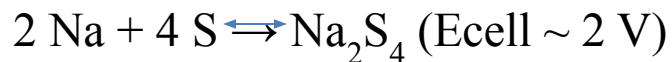
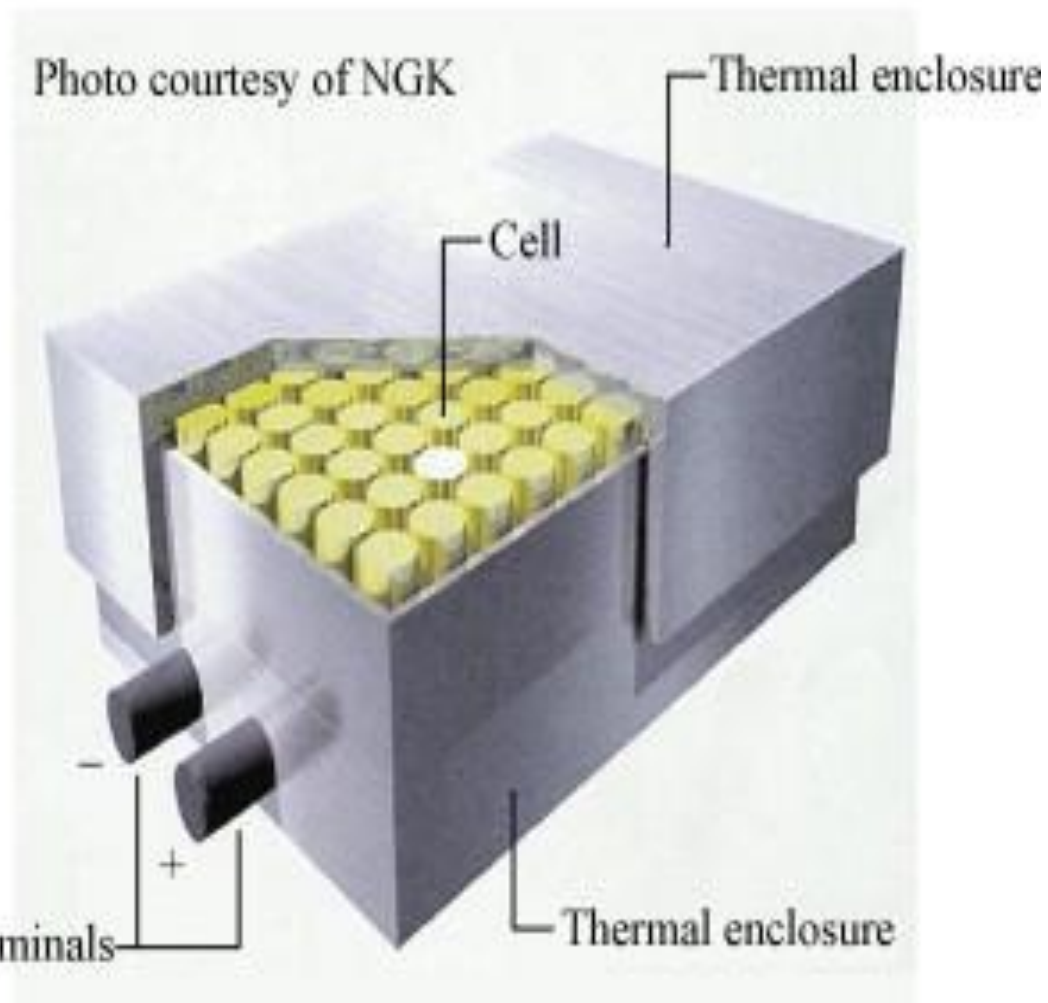
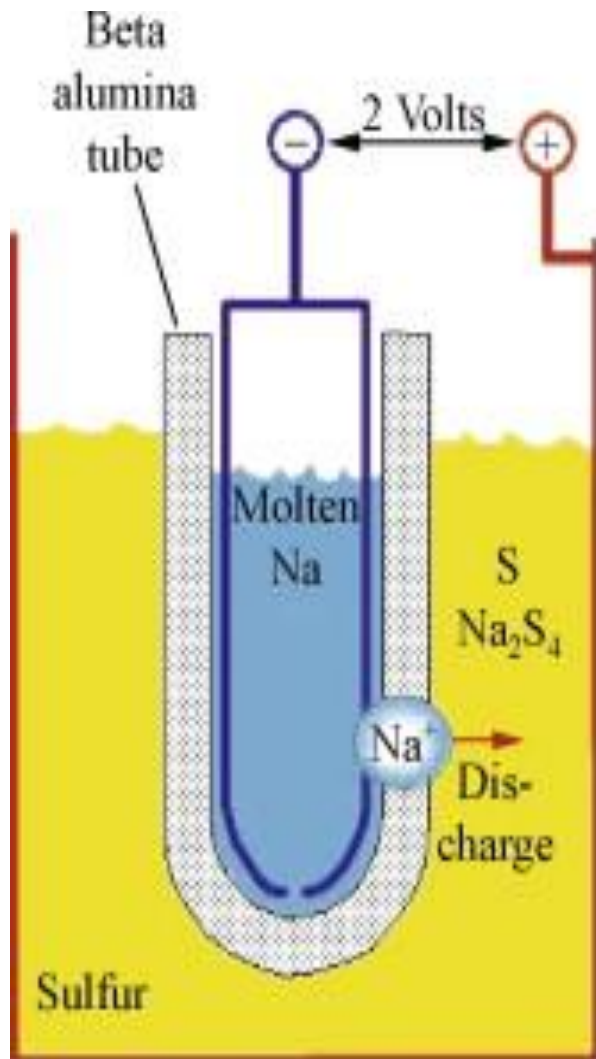


Advantages and disadvantages

- rechargeable sodium-ion batteries for energy storage
- Easier to recycle
- Low market prices.
- Capable of working at room temperature, Good efficiency.

- large ionic size of Na^+ which require more power to keep energy flowing.
- It takes several days to charge in case you forget to charge it full
- lower operating voltage
- lower energy density
- Need high temperature for optimal work

Sodium sulphur Battery

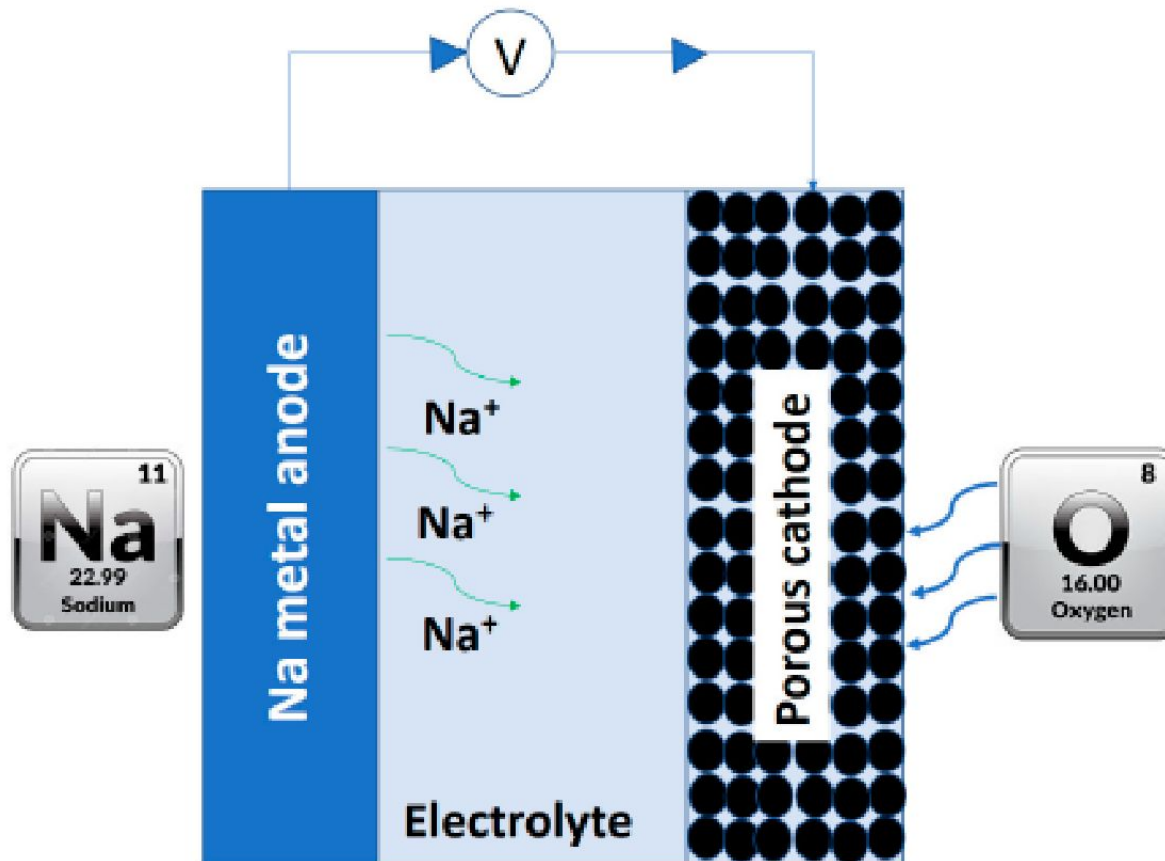




Sodium sulphur Battery

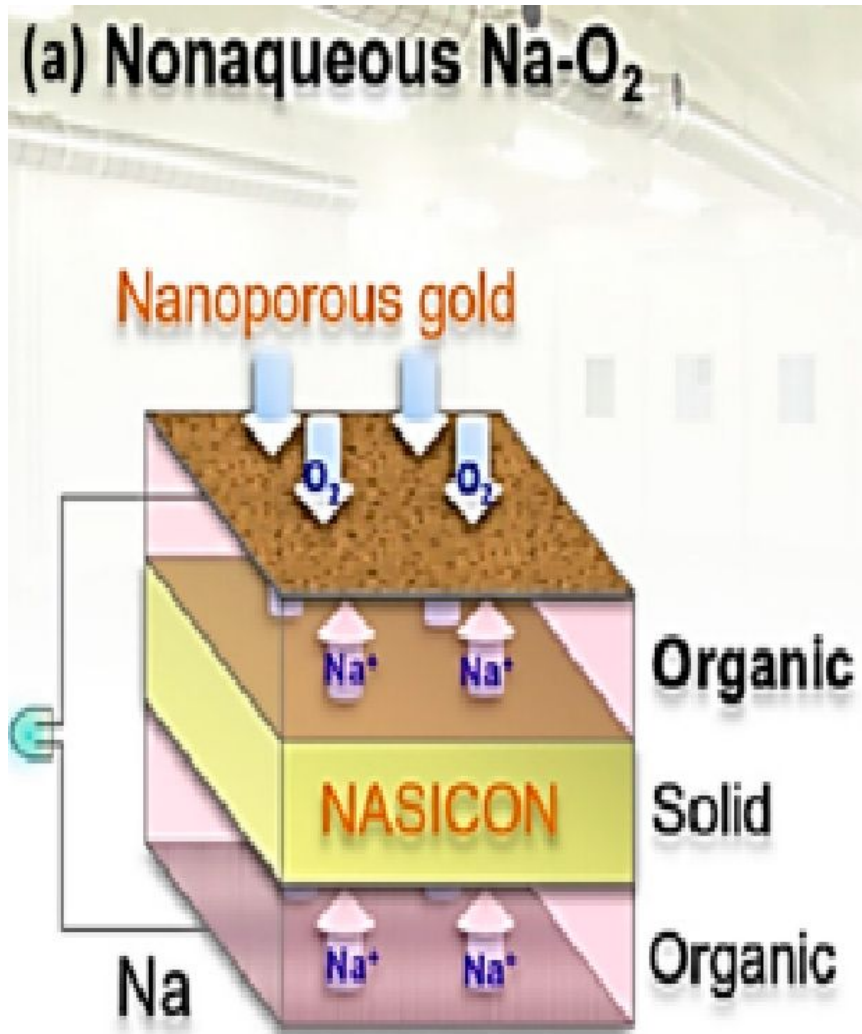
Advantages	Disadvantages
Low-cost potential: Inexpensive raw materials and sealed, no-maintenance configuration	Need to be operated above 300°C
High cycle life; liquid electrodes	Highly reactive nature of metallic sodium (part of the material used in construction), which is combustible when exposed to water
Good energy and power density: Low-density active materials, high cell voltage	Extra cost of constructing the enclosing structure to prevent leakage
Flexible operation: Cells functional over a wide range of conditions (rate, depth of discharge, temperature)	Stringent operation and maintenance requirements
High energy efficiency: 100% coulombic-efficient, reasonable resistance	
Insensitivity to ambient conditions: Sealed, high-temperature systems	
State-of-charge identification: Voltage rise and top-of-charge and end-of-discharge	

Sodium air Battery

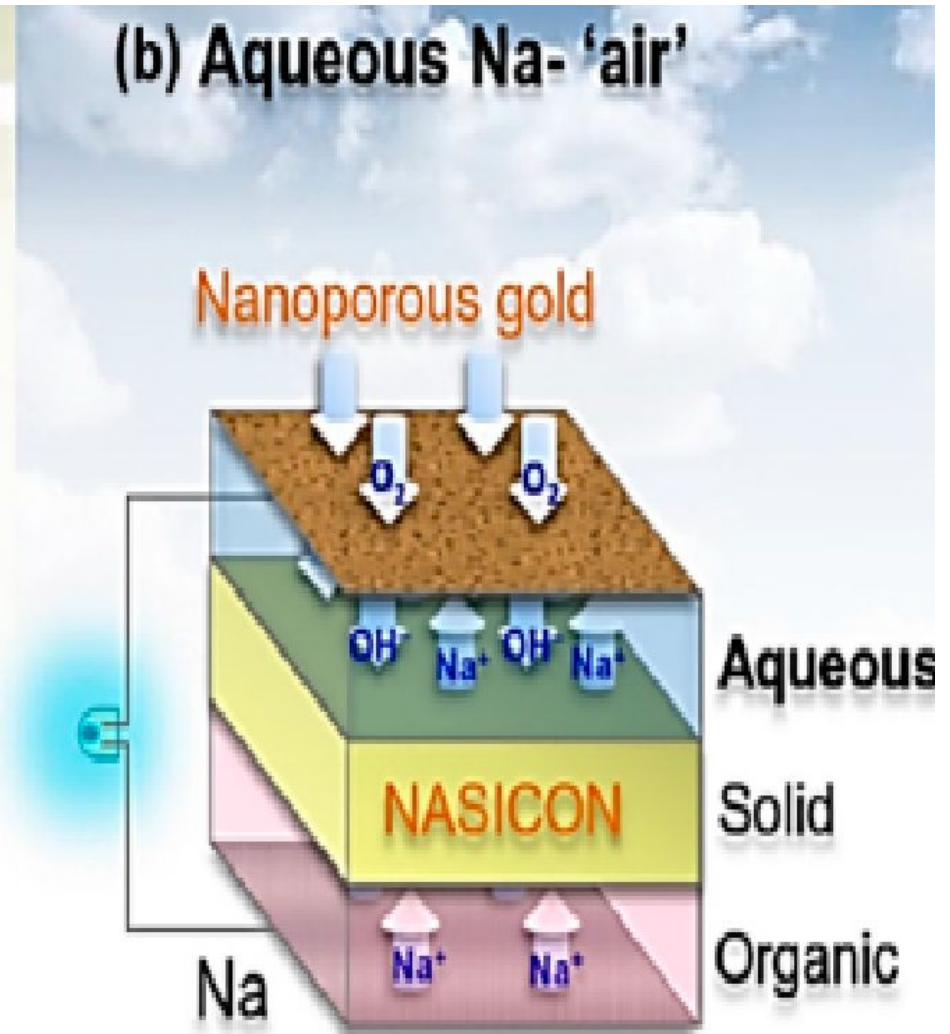


Types of Sodium air Battery

(a) Nonaqueous Na-O₂



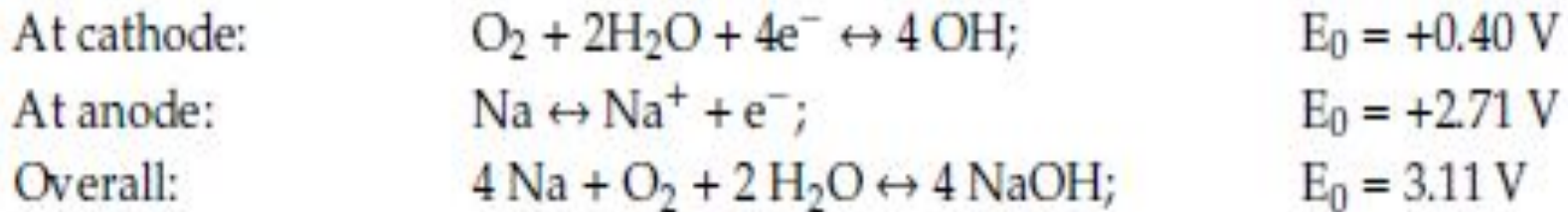
(b) Aqueous Na- 'air'



Sodium superionic conductor

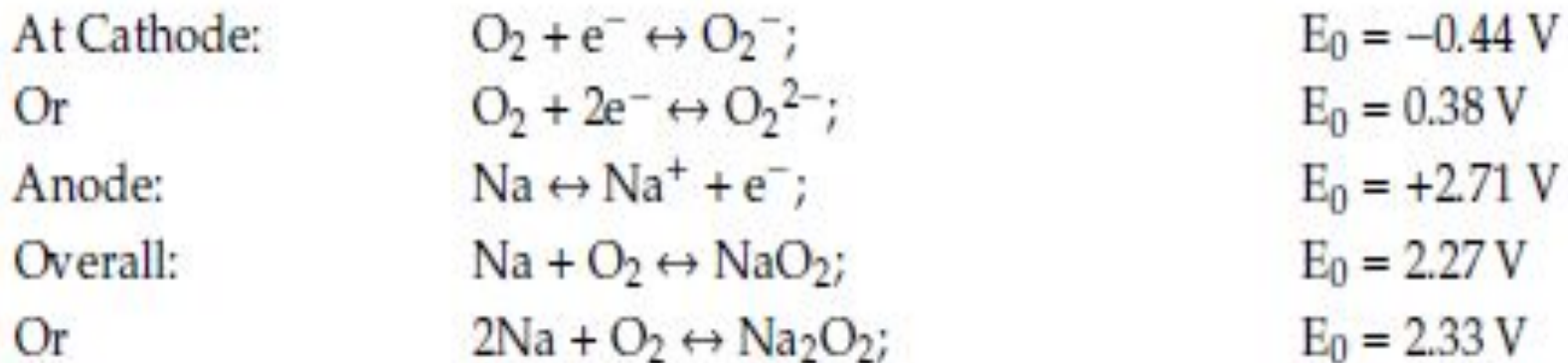


Electrochemical reactions in Sodium air Battery

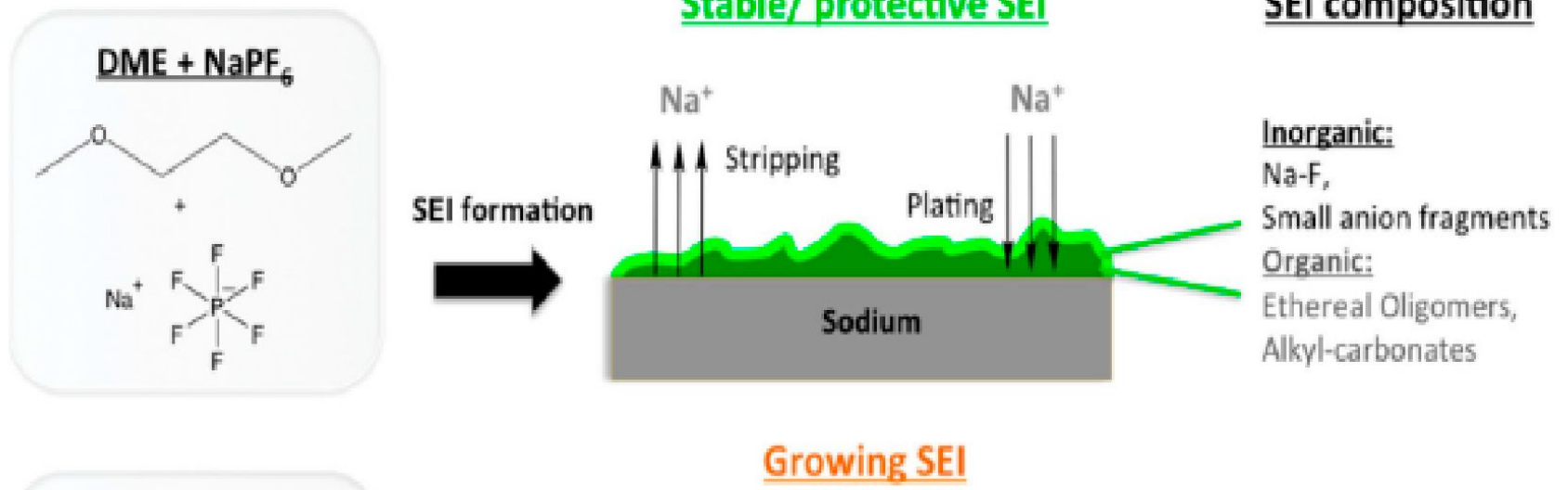


The standard cell voltage based on the above equations is 3.11 V.

However, for aprotic Na-air batteries, the reactions are given as follows



Sodium air Battery: Electrolytes





Advantages and disadvantages of Sodium air-battery

Advantages

- Na-air batteries have a theoretical energy density of $1,605 \text{ Wh kg}^{-1}$, which is significantly higher than that of Li-ion batteries
- Na-air battery demonstrated here is its very low overpotential, which is three or four times lower than for any Li-air or Na-air battery previously reported, resulting in fewer losses
- NaO_2 is chemically very difficult to synthesize. High temperatures, pressures and long reaction times are needed. But it can be prepared in battery easily in room temperature.

Major limitation

Na-air batteries can be charged and discharged several times, the capacity decreases after each cycle



Advantages and disadvantages of Sodium air-battery

Pros

Cheaper materials and simpler processing

Safer and less prone to overheating

Potential for large-scale energy storage

Cons

About 30 lower energy density than Li-ion batteries.

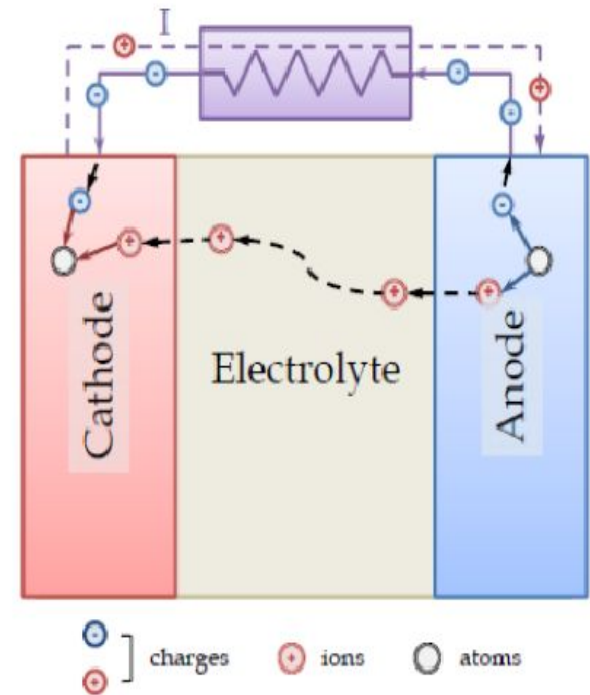
Larger and heavier than lithium batteries

Currently unsuitable for high-range electric vehicles

Magnesium Battery

Why Magnesium?

- **Mg batteries involve cycling Mg^{2+}**
 - Provides 2 electrons. Theoretical specific capacity: 2205 Ah/kg
- **Mg compounds are environmental stable “green” and non-toxic.** Mg alloys can be readily stabilized.
- **Mg as a raw materials is cheap and abundant**
 - Mg: ~ \$2700/ton
 - Li: ~ \$64,000/ton
 - Mg: ~ 13.9% earth's crust
 - Li: ~ 0.0007% earth's crust



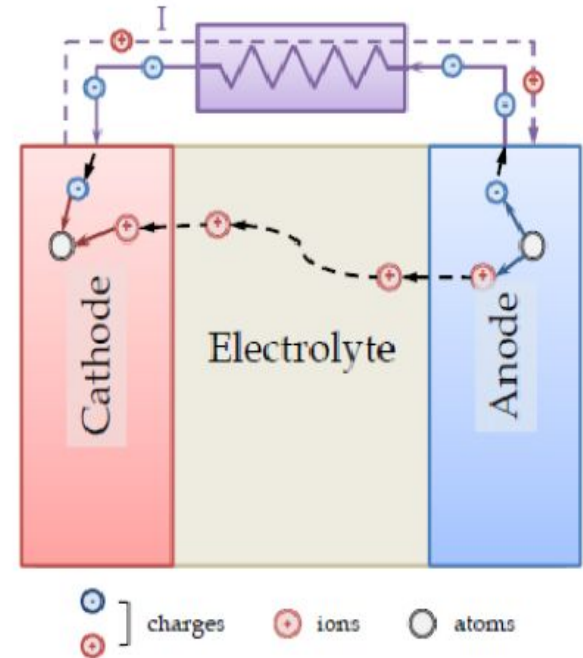
Systems with Proper Designs and Architectures:

- ***200-1000 Wh/kg***
- ***1-3 V***

Magnesium Battery

Key Challenges

- **Electrolytes**
 - Solvents: neither donate or accept protons
 - Solvents: Not allow the formation of passivating surface films onto Mg anode
- **Cathodes**
 - Difficulty in intercalation of Mg ions
 - Cycling of Mg systems is quite poor
 - Cycling is a key requirement for a rechargeable battery



Rechargeable Mg batteries recent work:

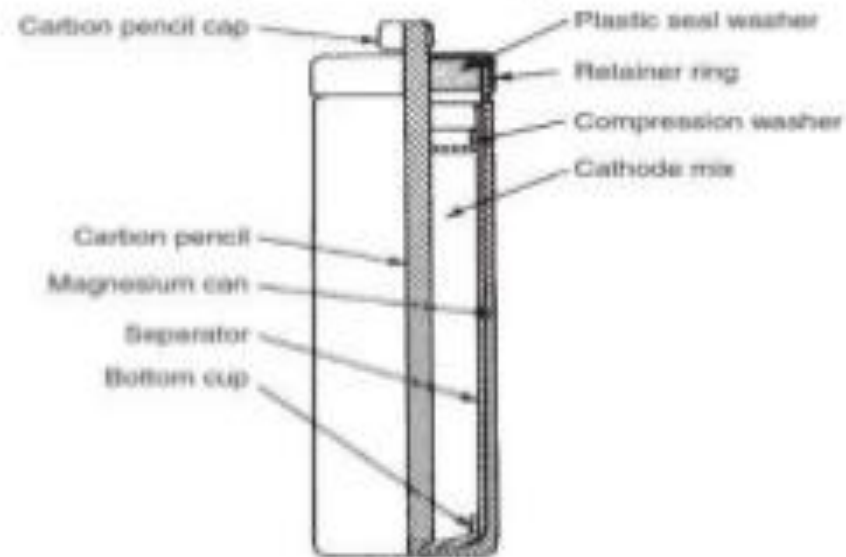
- **Electrolytes**
 - Aurbach et al ^{1,2}: Magnesium-organohaloaluminate salts
- **Cathodes**
 - Aurbach et al ^{1,2}: MgMo_3S_4
 - NuLi et al ³: $\text{Mg}_{1.03}\text{Mn}_{0.97}\text{SiO}_4$
- **Anodes**
 - Aurbach et al ^{1,2}: Mg foil, AZ31

- 1) D. Aurbach et al, *Advanced Materials*, 2007, v19, p 4260
- 2) E. Levi et al, *Chem. Materials*, 2010., v22, p. 860
- 3) Y. NuLi et al. *Chem. Comm.* 2010, v46., p. 3794.

Construction and working

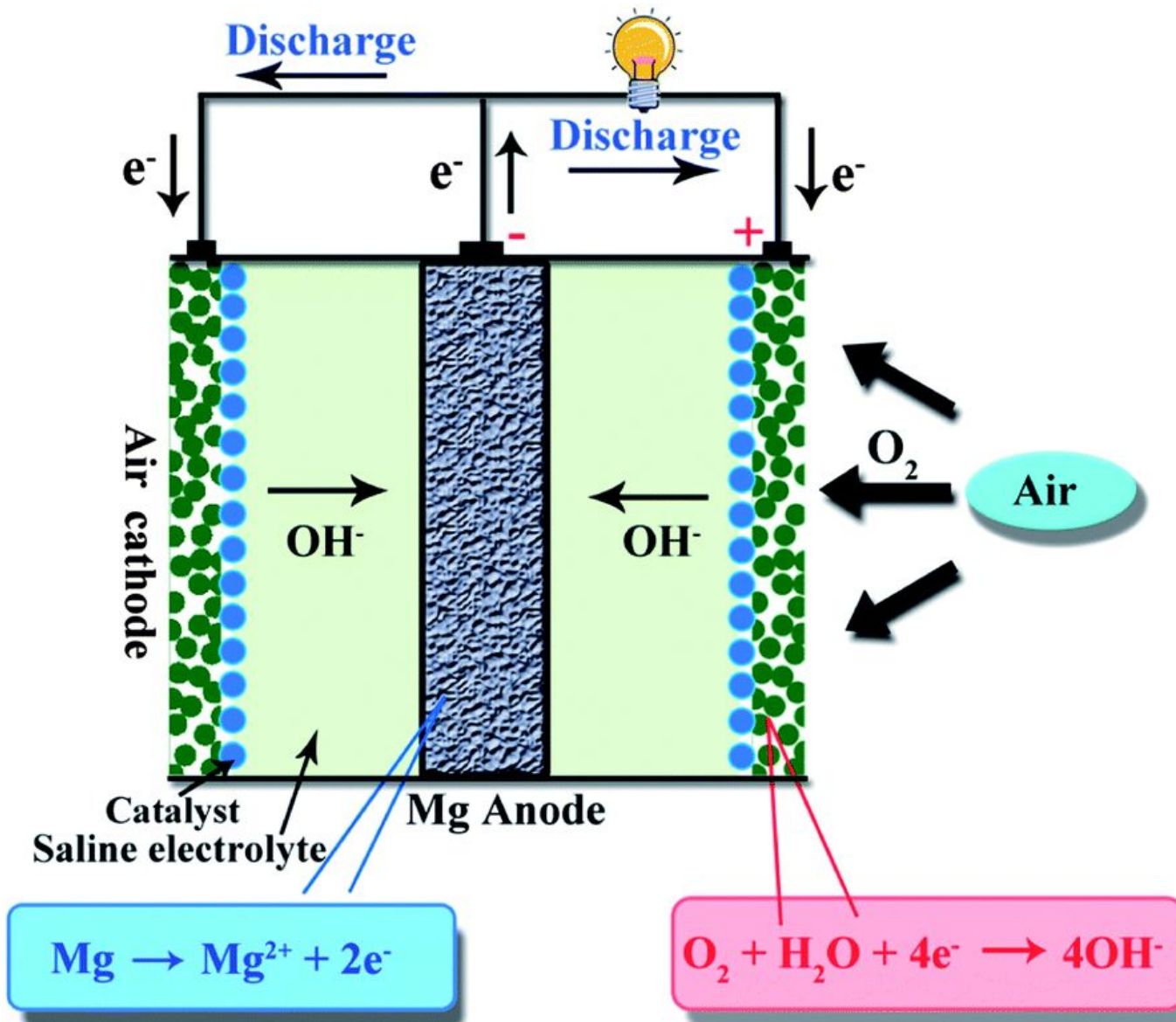
- ▶ Twice the service life or capacity of zinc battery
- ▶ Disadvantages – voltage delay and parasitic corrosion
- ▶ Potential > 2.8V, but 1.1V is achieved
- ▶ Battery chemistry, $\text{Mg} + 2 \text{MnO}_2 + \text{H}_2\text{O} \rightarrow \text{Mn}_2\text{O}_3 + \text{Mg}(\text{OH})_2$

Figure represents
Magnesium batteries

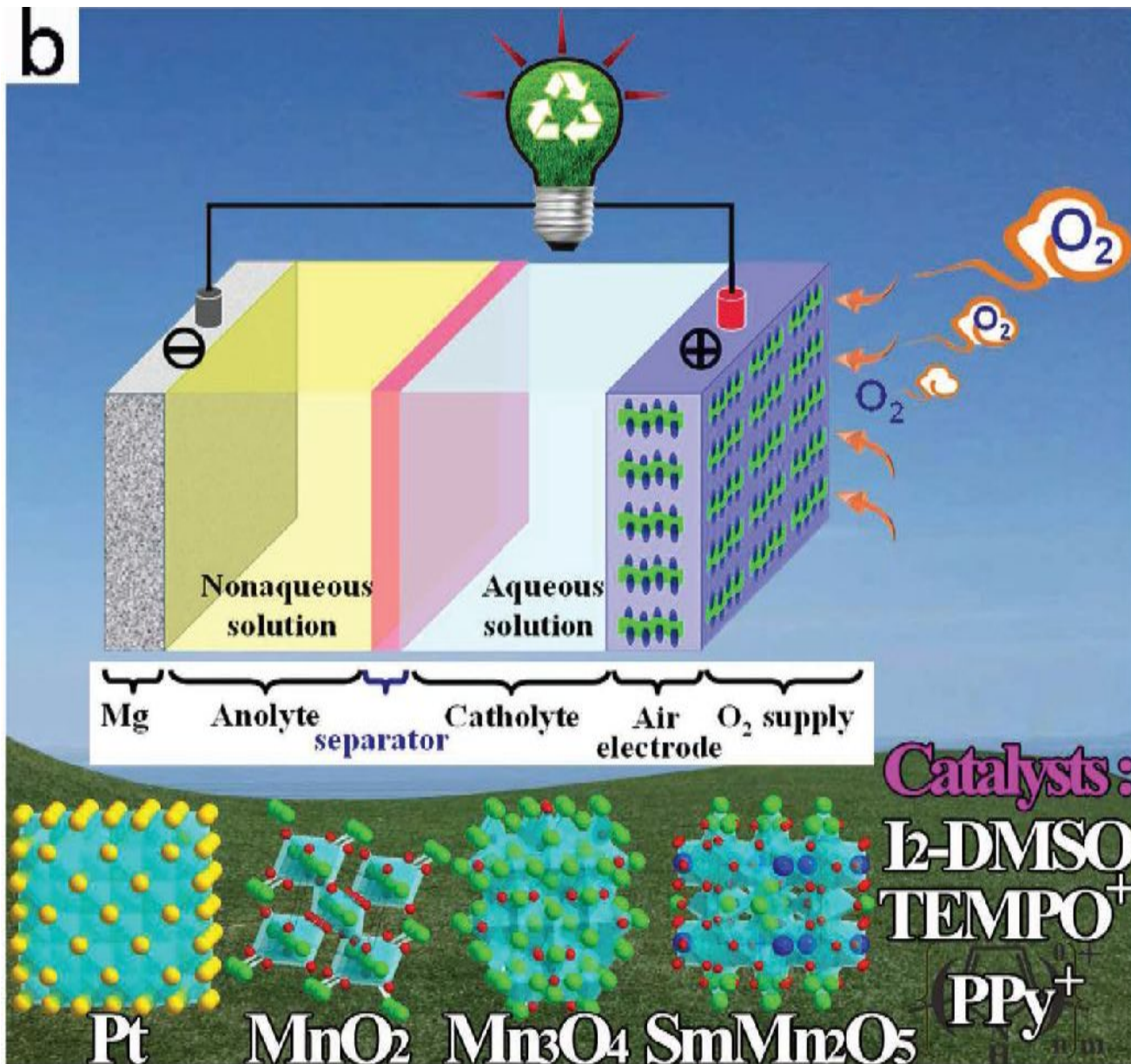


[2]

Construction and working of Mg-Air battery



Construction and working of Mg-Air battery





Anode and cathode materials used in of Mg-Air battery

Anode materials

Simple Mg

Commercial Mg

Mg

Nano/mesoscale Mg

Mg

Mg-alloy

AZ31, AZ61, AZ91^a

Mg/Al/Zn

AM50, AM60, MA8M06^b

Mg/Al/Mn

Mg-Li alloys

Mg/Li

Cathode materials

Carbonaceous cathode

Graphite

Transition metal oxides

Metallic complexes

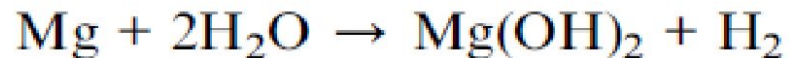
Nobel metal like platinum



Construction and working of Mg-Air battery

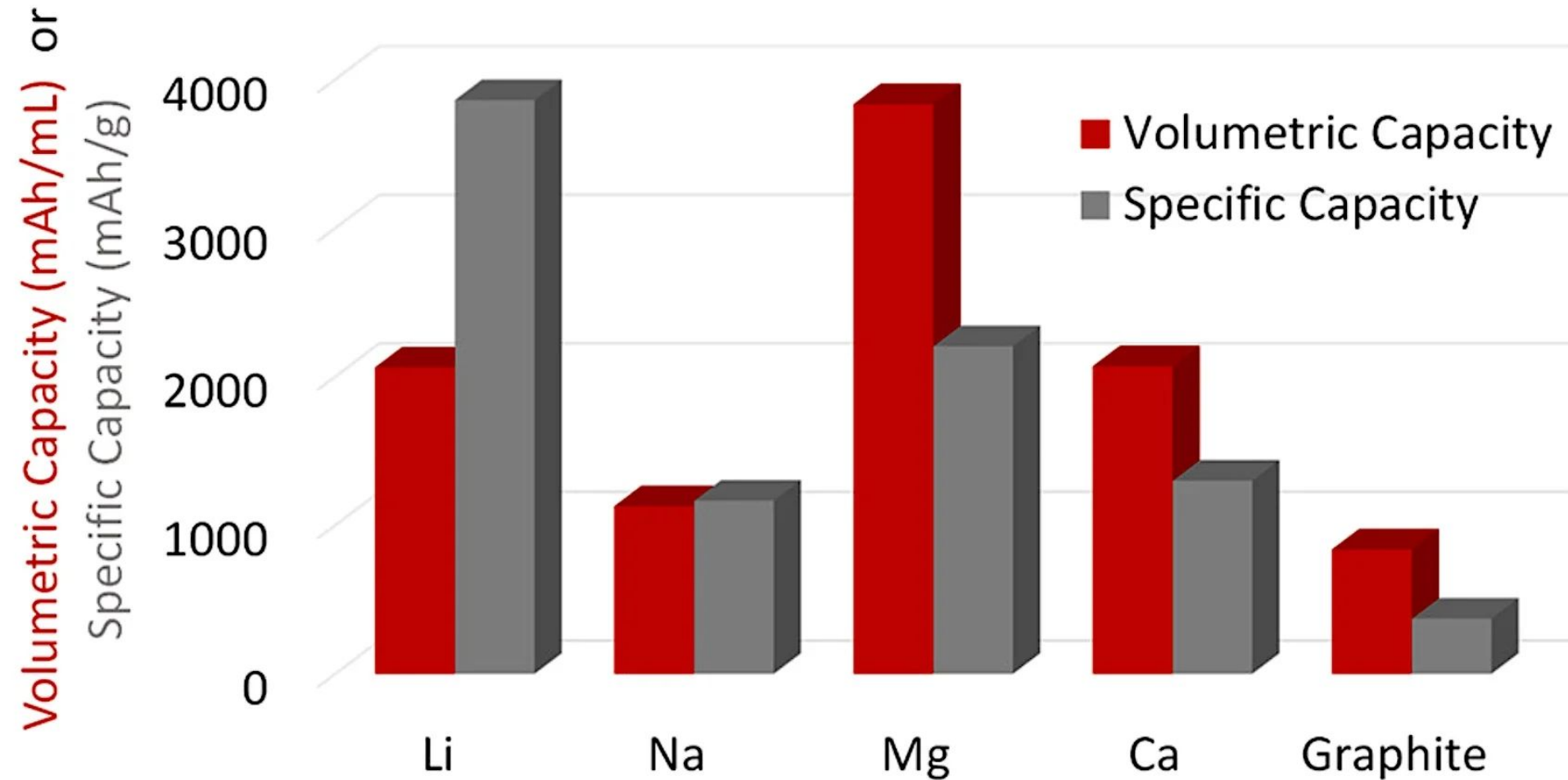


The overall corrosion reaction can be written as:



Comparison between Mg air and Li-air battery

Theoretical Capacities for Different Battery Materials





Comparison between Mg air and Li-air battery

- High energy density
- Safe
- Inexpensive
- Not been widely used - self-discharge in neutral solution
- Reaction mechanism of magnesium alloy anode
- Effects of different additives on performance of Mg alloy in solution
- Approach energy density of fuel cells
- Single use and rechargeable lithium metal-air – could power Evs
- Theoretically: max energy density 5,000+ watt-hours per kilogram
- Lower self discharge rate and longer shelf life



Mg air battery-Pros and cons

Pros

High theoretical energy density

Non-flammable and safer operation

Abundance of magnesium reduces costs

Lower environmental footprint than lithium

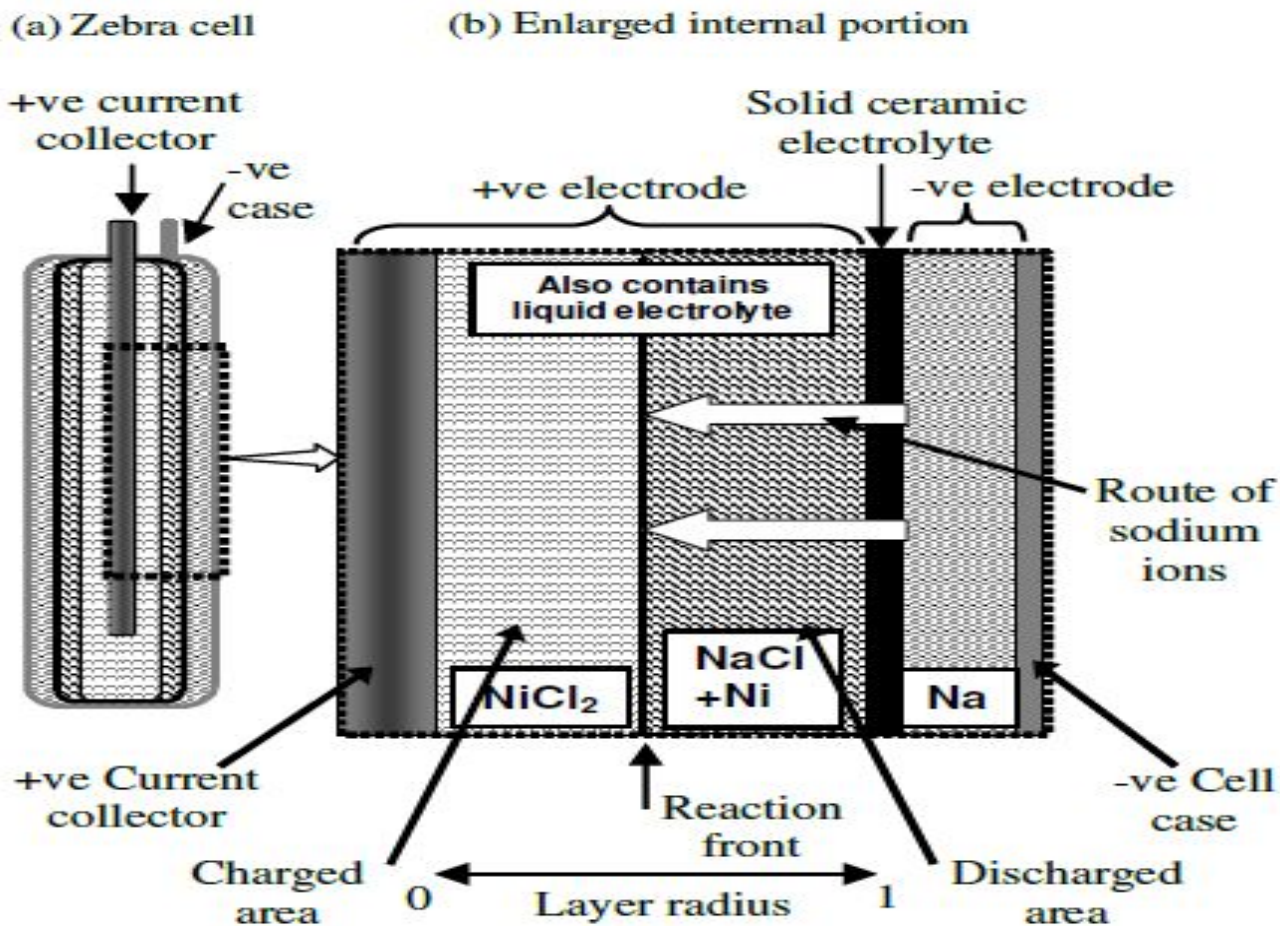
Cons

Limited availability of efficient electrolytes

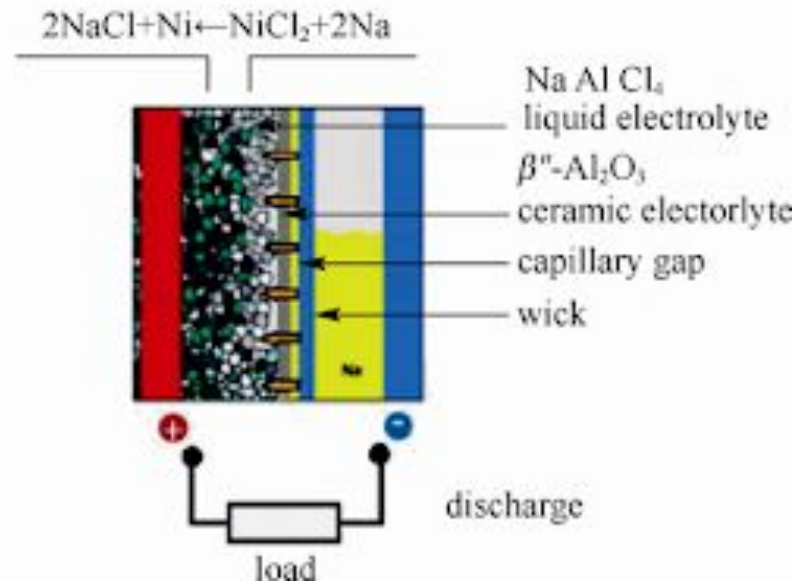
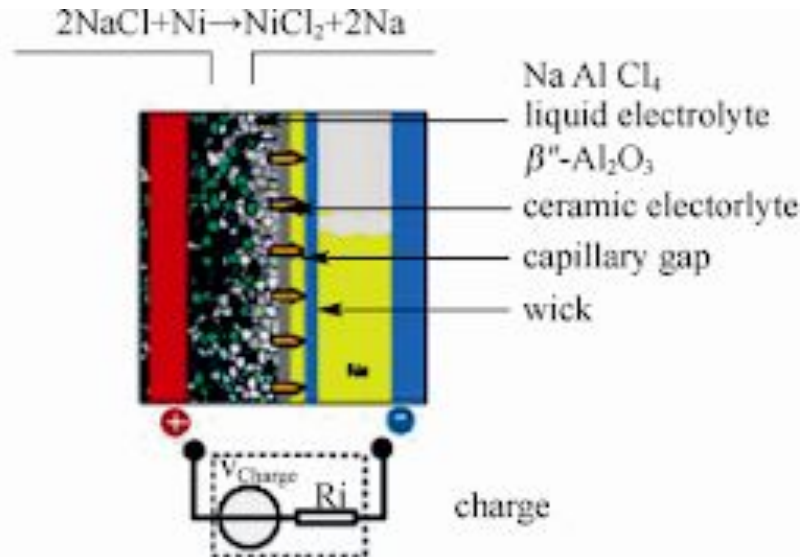
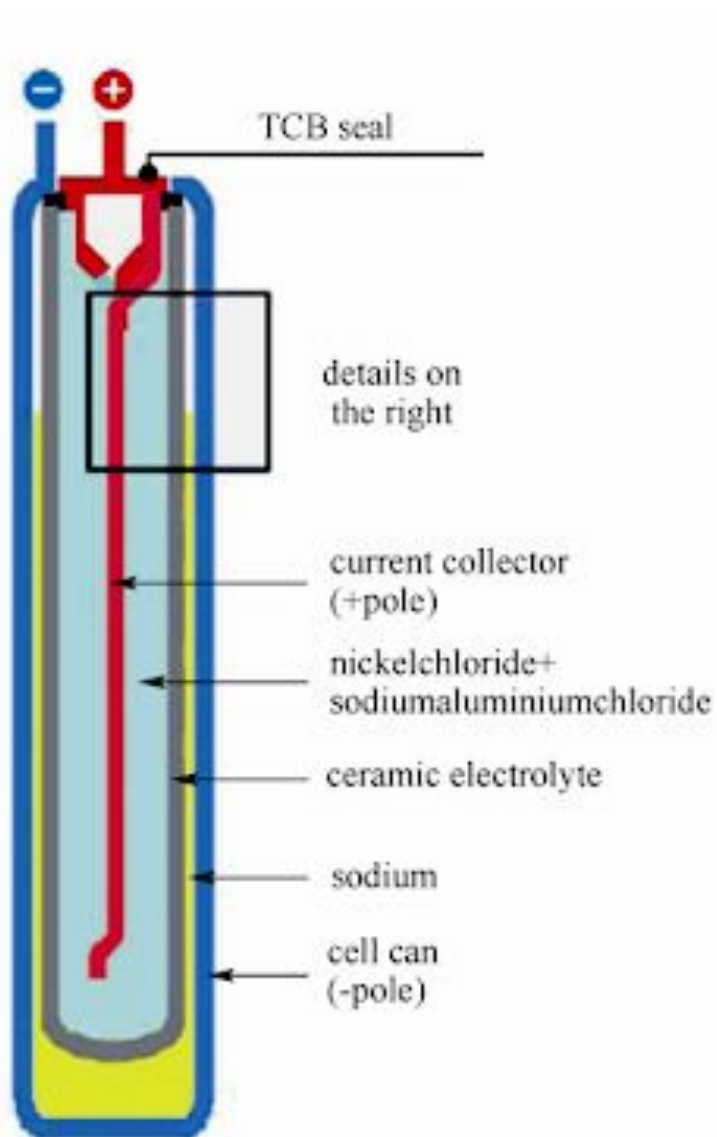
High internal resistance impacts energy efficiency

Technology is still in the research stage

Zebra battery



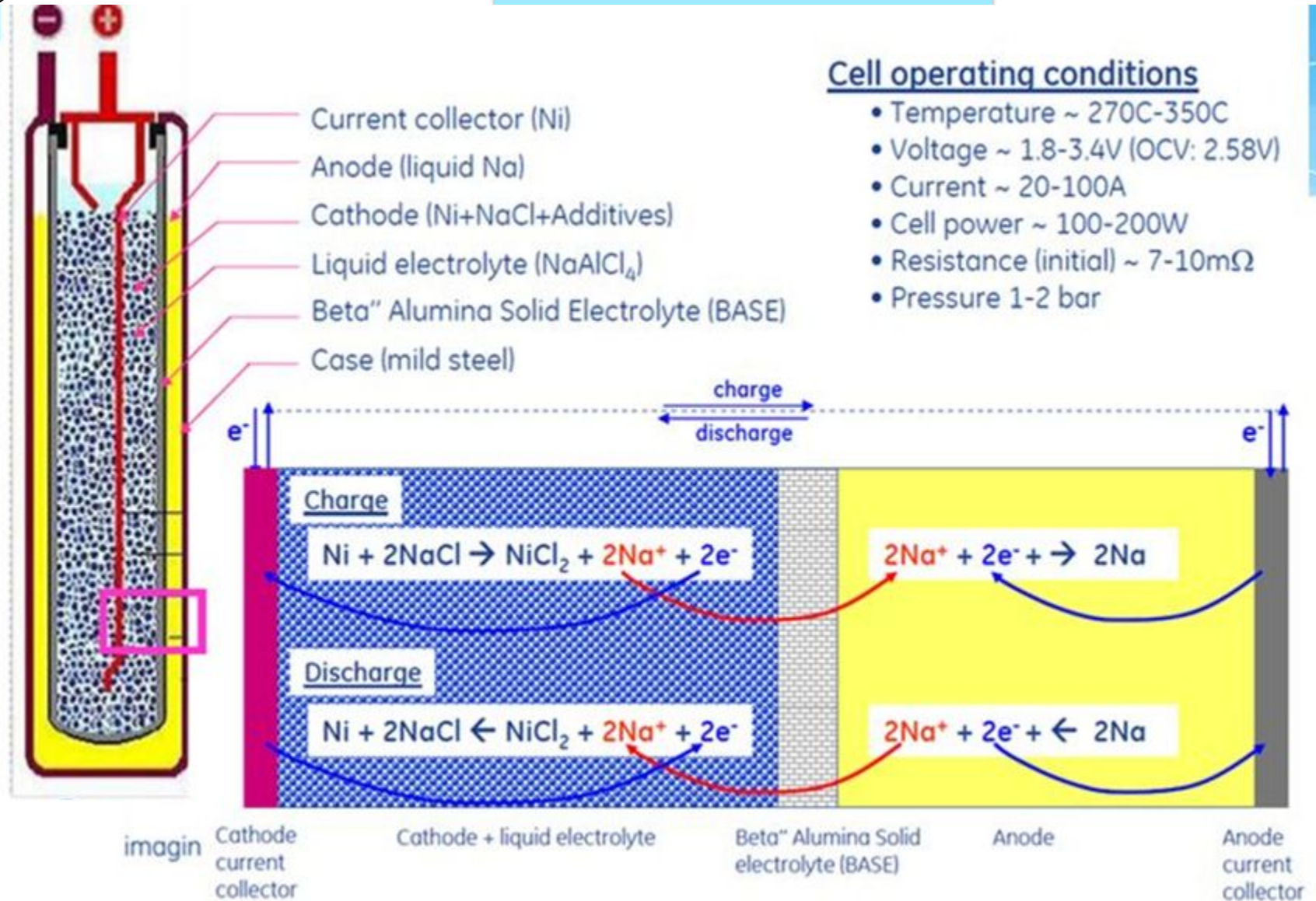
Zebra battery



Zebra battery

Cell operating conditions

- Temperature ~ 270C-350C
- Voltage ~ 1.8-3.4V (OCV: 2.58V)
- Current ~ 20-100A
- Cell power ~ 100-200W
- Resistance (initial) ~ 7-10mΩ
- Pressure 1-2 bar



Zebra battery

Anode (-): Na metal

Ni- Cu composite Current Collector

Electrolyte { Solid electrolyte: β "-Alumina ($\geq 0.2 \Omega^{-1}\text{cm}^{-1}$ at 260°C)
Liquid electrolyte: NaAlCl_4 ($0.6 \Omega^{-1}\text{cm}^{-1}$ at 250°C)

Cathode (+): Transition metal chloride
+ Excess metal



2.35 V @ 250°C

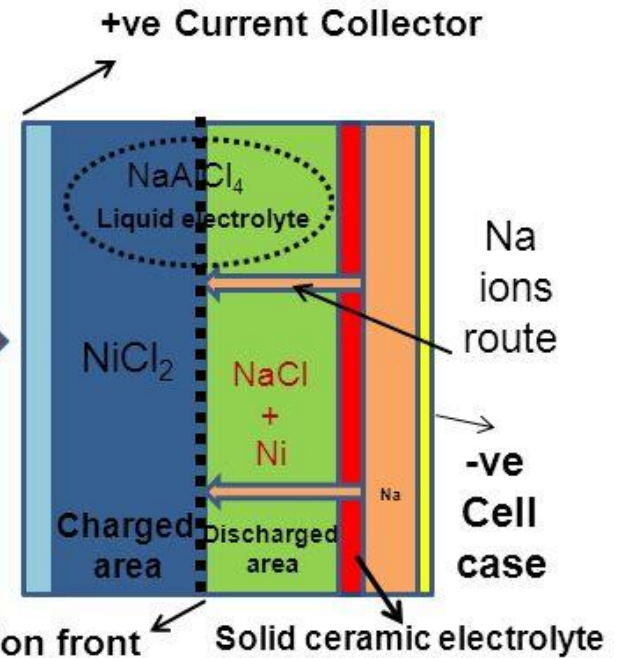
($200-300^\circ\text{C}$)

2.58 V @ 300°C

($200-400^\circ\text{C}$)

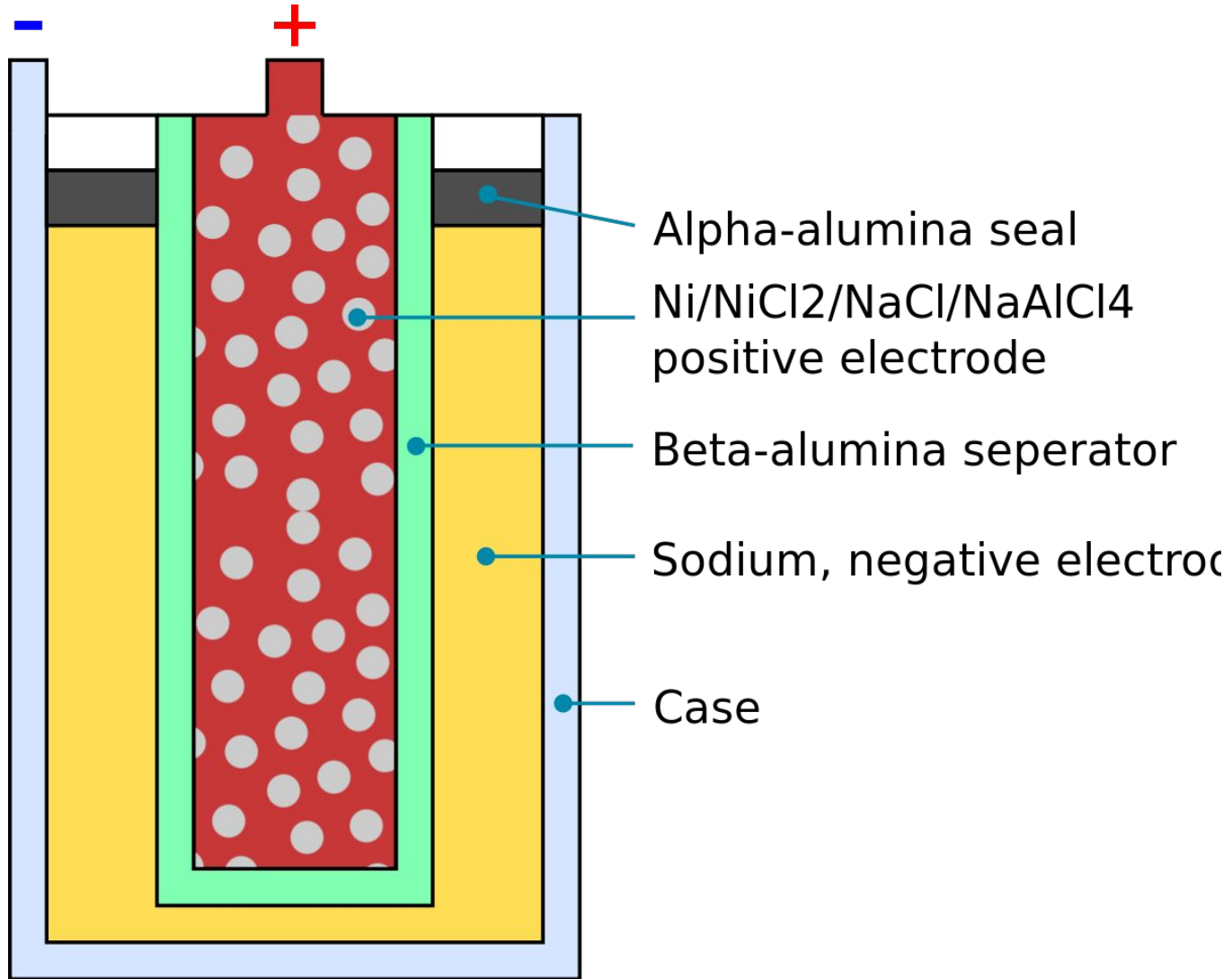


3



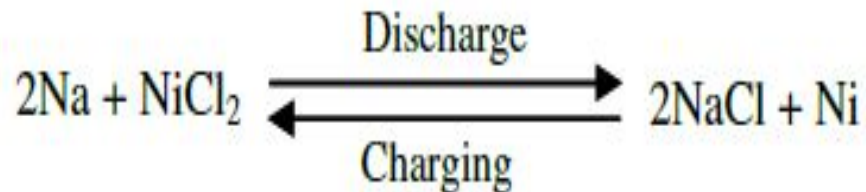
J.L. Sudworth, J. Pow. Sour., 100 (2001)

Zebra battery





Zebra battery



Characteristics

- High power, high capacity cells suitable for electric traction applications.
- High Temperature Battery operating at 270°C
- Sodium Nickel Chloride (NaNiCl) chemistry giving a nominal operation cell voltage of 2.58 Volts



Zebra battery: Advantages

- High energy density (5 times higher than Lead acid)
- Large cells (up to 500Ah) possible
- Cycle life better than 1000 cycles
- Tolerant of short circuits
- Safer than Sodium Sulphur cells
- Typical cell failure is short circuit which does not cause complete failure of the battery.
- Low cost materials



Zebra battery: Limitations

- Suitable for large capacity batteries only ($> 20\text{KWh}$)
- Limited range of available sizes and capacities. (Large multi-cell blocks)
- High internal resistance
- Molten sodium electrode
- High operating temperature.
- Preheating needed to get battery up to the 270°C operating temperature. (Up to 24 hours from cold)
- Uses 14% of its own capacity per day to maintain temperature when not in use.
- Thermal management needed



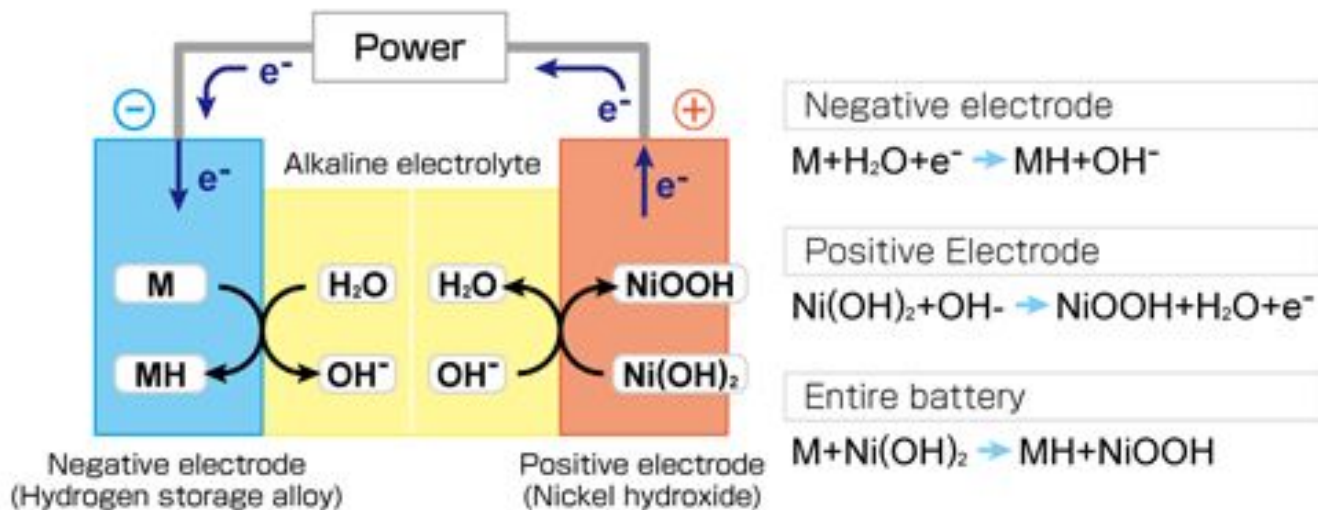
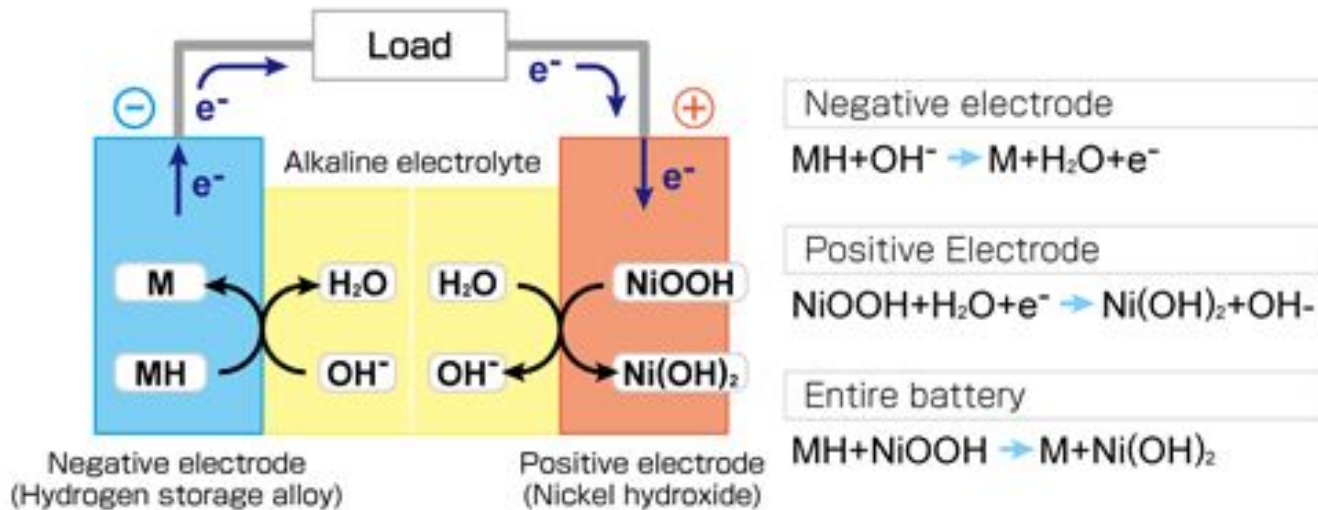
Comparison of Mg batteries

BATTERY	ZEBRA	NaS	Mg
Anode	Molten metallic Na	Molten Metallic Na	Solid Mg alloy
Cathode	NiCl_2 or NaAlCl_4	Molten S or Na_2S_x	$\text{Mg}_{1.03}\text{Mn}_{0.97}\text{SiO}_4$
Electrolyte	β - Al_2O_3 solid electrolyte (BASE)	β - Al_2O_3 solid electrolyte (BASE) membrane	$\text{Mg}(\text{AlCl}_2\text{BuEt})_2/\text{THF}$ (0.25 mol L ⁻¹)
Cell Voltage	2.58 V	2.1 V	2.1 V
Specific Capacity*	305 Ah/kg	377 Ah/kg	315 Ah/kg**
Specific Energy*	100 Wh/kg	110 Wh/kg	~500 Wh/kg
Operating Temperature	High 270°C	High 350°C	Room Temp

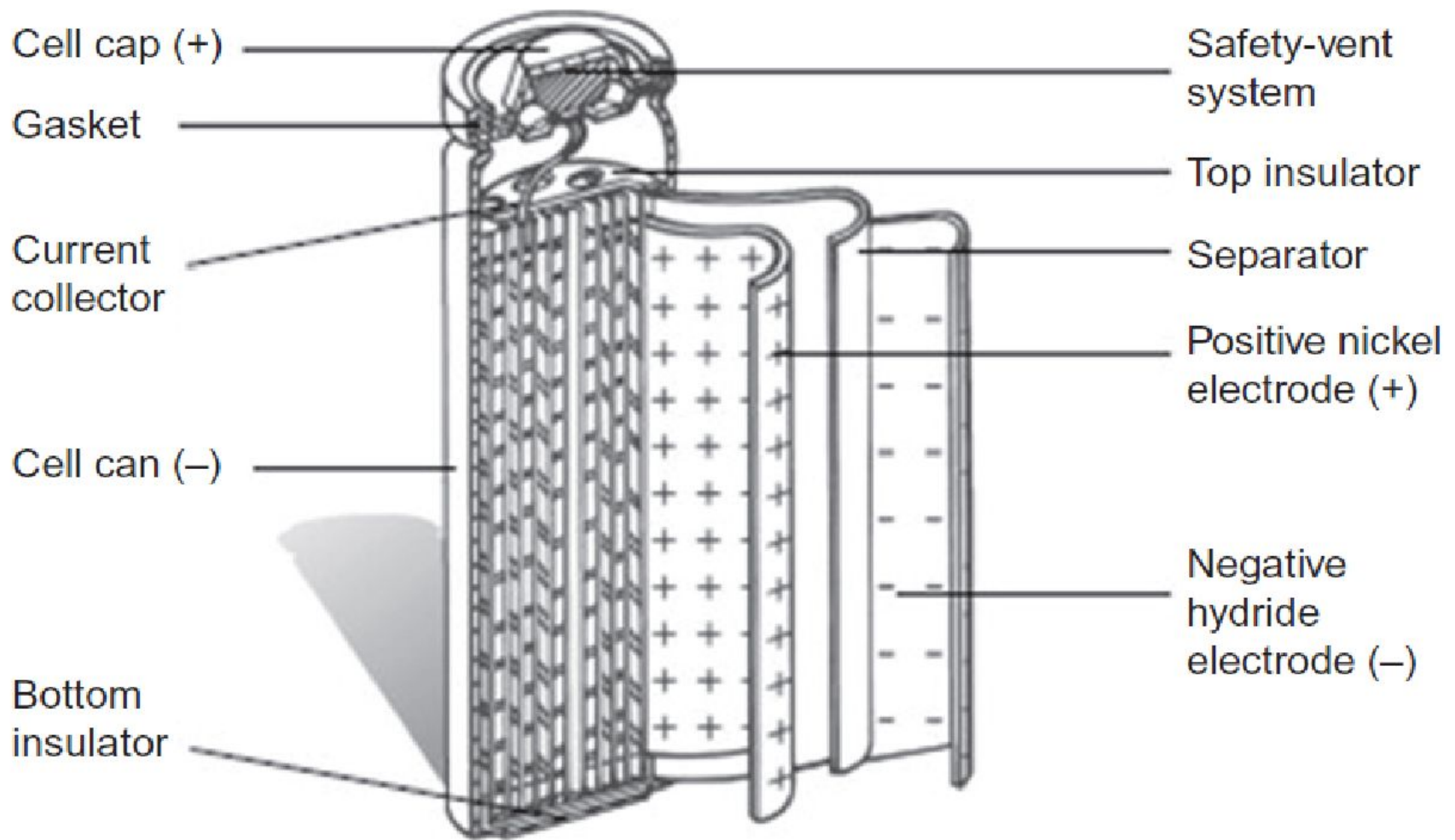
*Theoretical

**based on cathode

Ni MH Battery

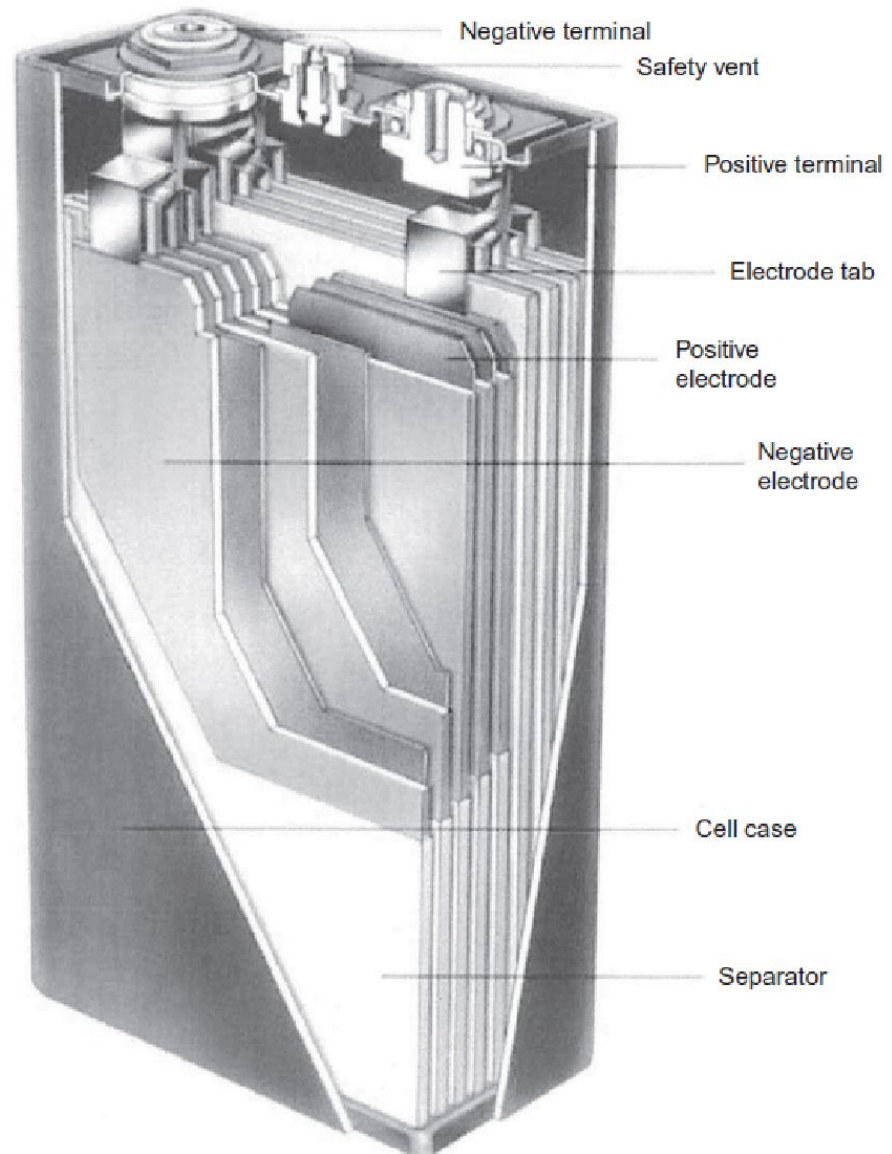


Ni MH Battery

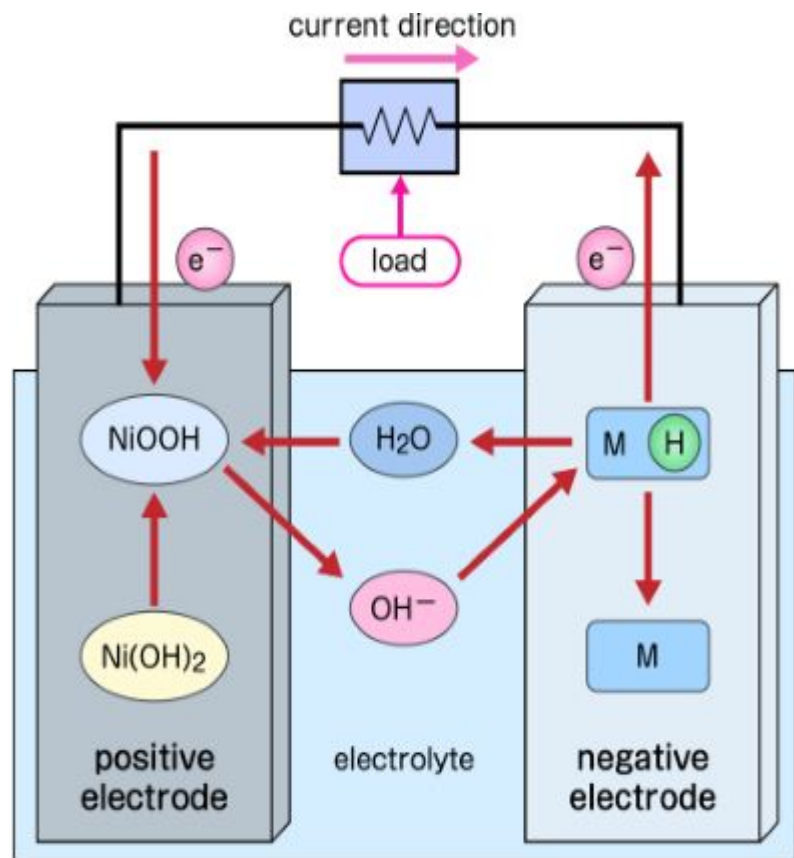


Ni MH Battery

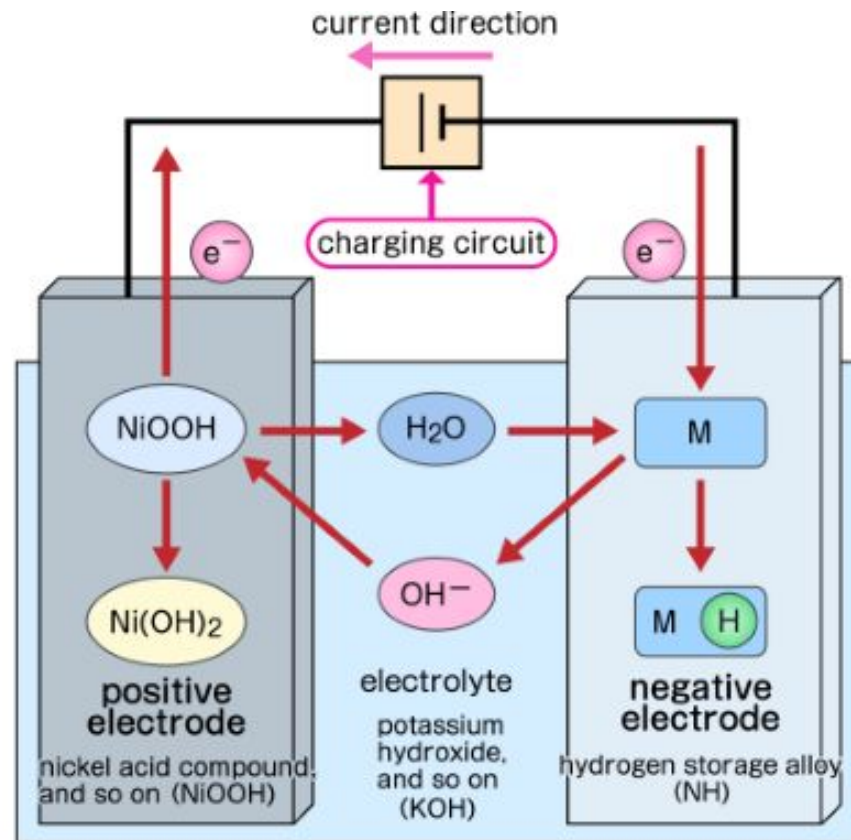
- Negative electrode: Metal Hydride such as AB₂ (A=titanium and/or vanadium, B= zirconium or nickel, modified with chromium, cobalt, iron, and/or manganese) or AB₅ (A=rare earth mixture of lanthanum, cerium, neodymium, praseodymium, B=nickel, cobalt, manganese, and/or aluminum)
- Positive electrode: nickel oxyhydroxide (NiO(OH))
- Electrolyte: Potassium hydroxide (KOH)



Ni MH Battery



(b) in case of discharge



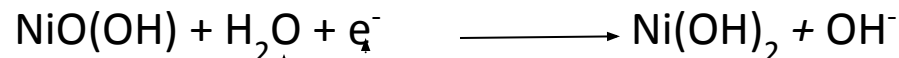
(a) in case of charge

Ni MH Battery

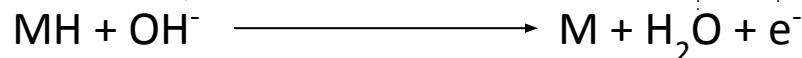
The nickel–metal hydride battery makes use of hydrogen for the positive electrode. This hydrogen is stored in alloy (i.e., metal hydride). The reactions of the battery during charging and discharging are illustrated in equations

- Chemical reaction (discharge)

- Positive electrode



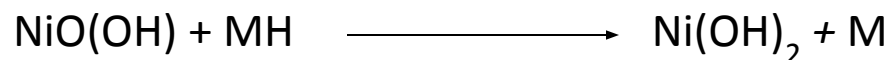
- Negative electrode



Through electrolyte

Through load

- Overall



The electrolyte is not affected because it does not participate in the reaction.

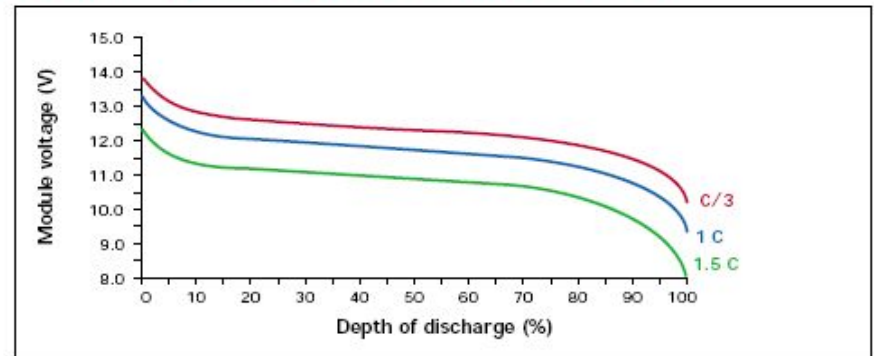
Ni MH Battery

- It is not advisable to charge Ni-MH batteries with a constant-voltage method. Ni-MH batteries do not accept well a high initial charging current.
- Float voltage is about 1.4 V
- Minimum voltage is about 1 V.

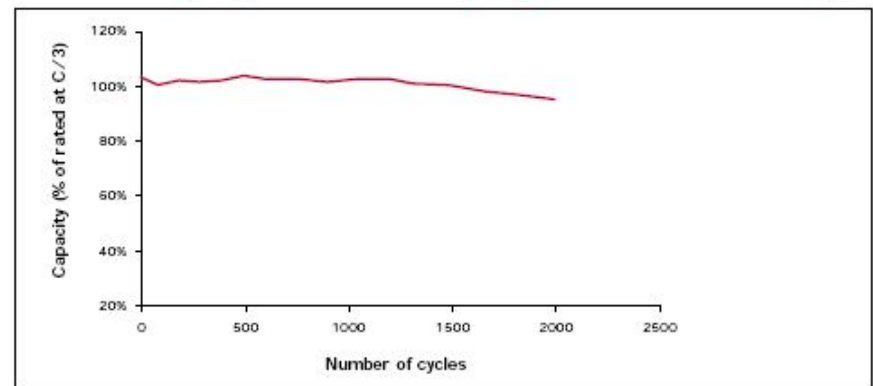


Cobasys Nigen battery

NHE module - Discharge at various rates at +23°C (+73.5°F)



NHE module - Cycling at 80% DOD* - Capacity at C/3 rate after full charge

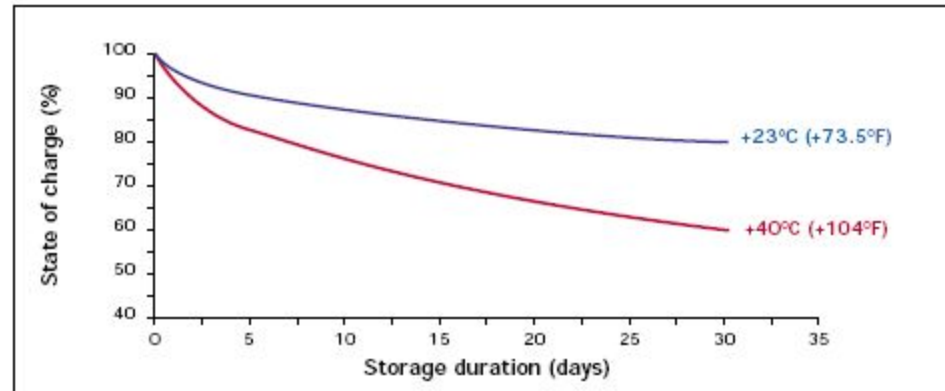


Saft NHE module battery

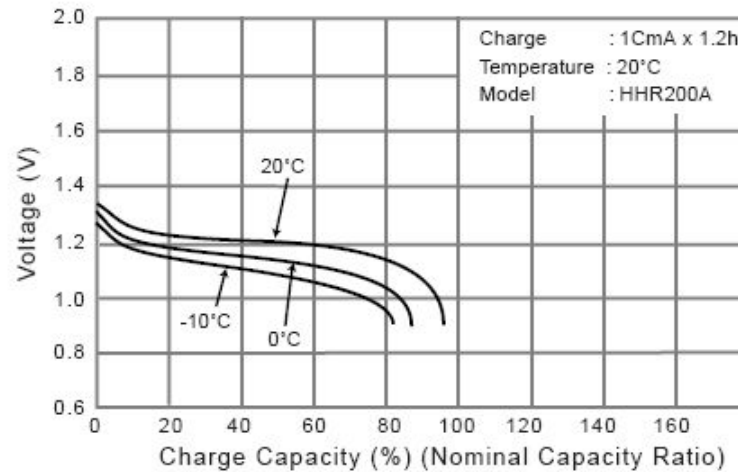
Ni MH Battery

- Effects of temperature:

NHE module - Charge retention at different temperatures



Soft NHE module battery



http://www.panasonic.com/industrial/battery/oem/images/pdf/panasonic_nimh_overview.pdf



Ni MH Battery

- Advantages:
 - Less sensitive to high temperatures than Li-ion and Lead-acid
 - Handle abuse (overcharge or over-discharge better than Li-ion bat
- Disadvantages:
 - More cells in series are need to achieve some given voltage.
 - Cost

	Lead-Acid	Ni-Cd	Ni-MH	Li-ion
Cell voltage (V)	2	1.2	1.2	3.6
Specific energy (Wh/kg)	1-60	20-55	1-80	3-100
Specific power (W/kg)	< 300	150 – 300	< 200	100 – 1000
Energy density (kWh/m ³)	25-60	25	70-100	80-200
Power density (MW/m ³)	< 0.6	0.125	1.5 – 4	0.4 – 2
Maximum cycles	200-700	500-1000	600-1000	3000
Discharge time range	> 1 min	1 min-8 hr	> 1 min	10 s-1 h
Cost (\$/kWh)	125	600	540	600
Cost (\$/kW)	200	600	1000	1100
Efficiency (%)	75 - 90	75	81	99



Ni MH Battery

Advantages

Energy density, which can be translated into either long run times or reduction in the space needed for the battery

Elimination of the constraints imposed on battery manufacture, usage, and disposal because of concerns over cadmium toxicity

Simplified incorporation into products currently using nickel-cadmium batteries because of the many design similarities between the two chemistries

Greater service advantage over other primary battery types at extreme low-temperature operation (-20°C)

Disadvantages

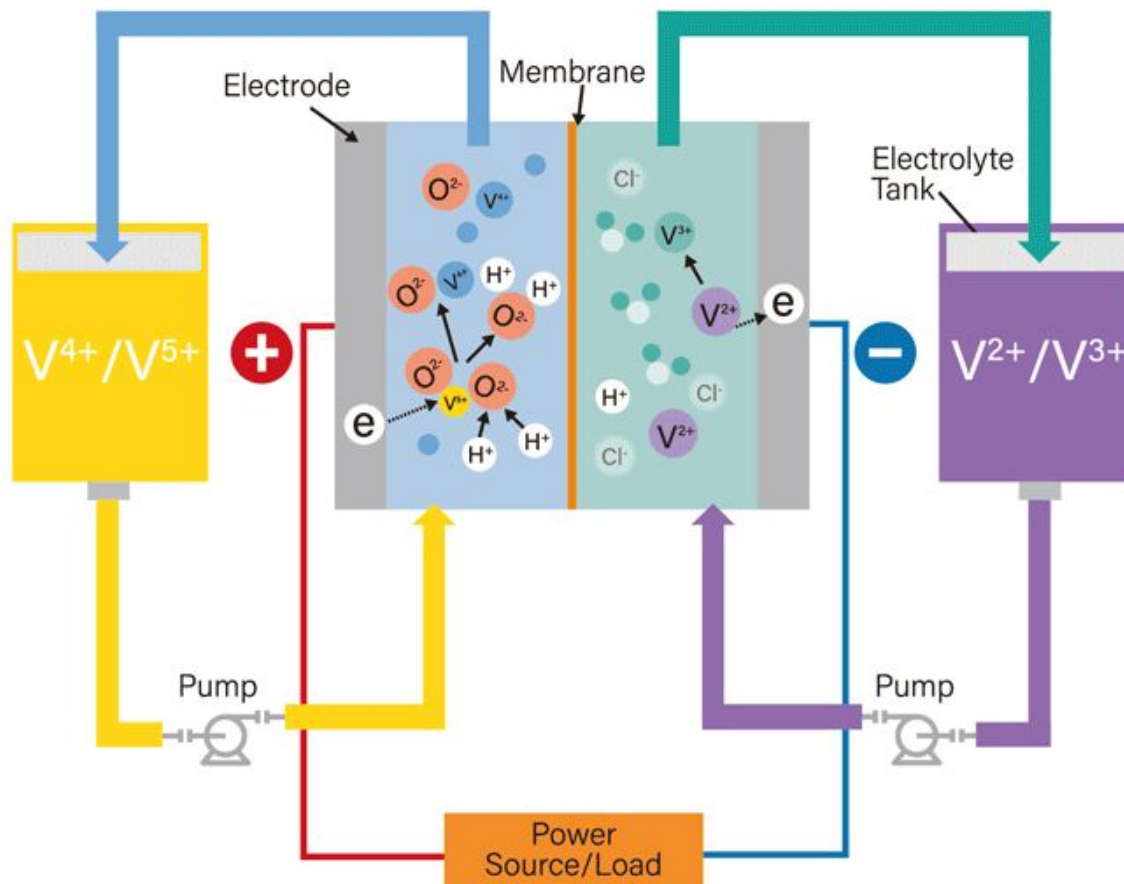
Limited service life: If repeatedly deep-cycled, especially at high load currents, performance starts to deteriorate after 200–300 cycles. Shallow, rather than deep, discharge cycles are preferred.

Limited discharge current: Although a Ni-MH battery is capable of delivering high discharge currents, repeated discharge with high load currents reduces the battery's cycle life. Best results are achieved with load currents of 0.2–0.5 C (one-fifth to one-half of the rated capacity).

Need for a more complex charge algorithm: The Ni-MH generates more heat during charge and requires a longer charge time than the Ni-Cd. The trickle charge is critical and must be controlled carefully.

High self-discharge: The Ni-MH has about 50% higher self-discharge compared with the Ni-Cd. New chemical additives improve the self-discharge, but at the expense of lower energy density.

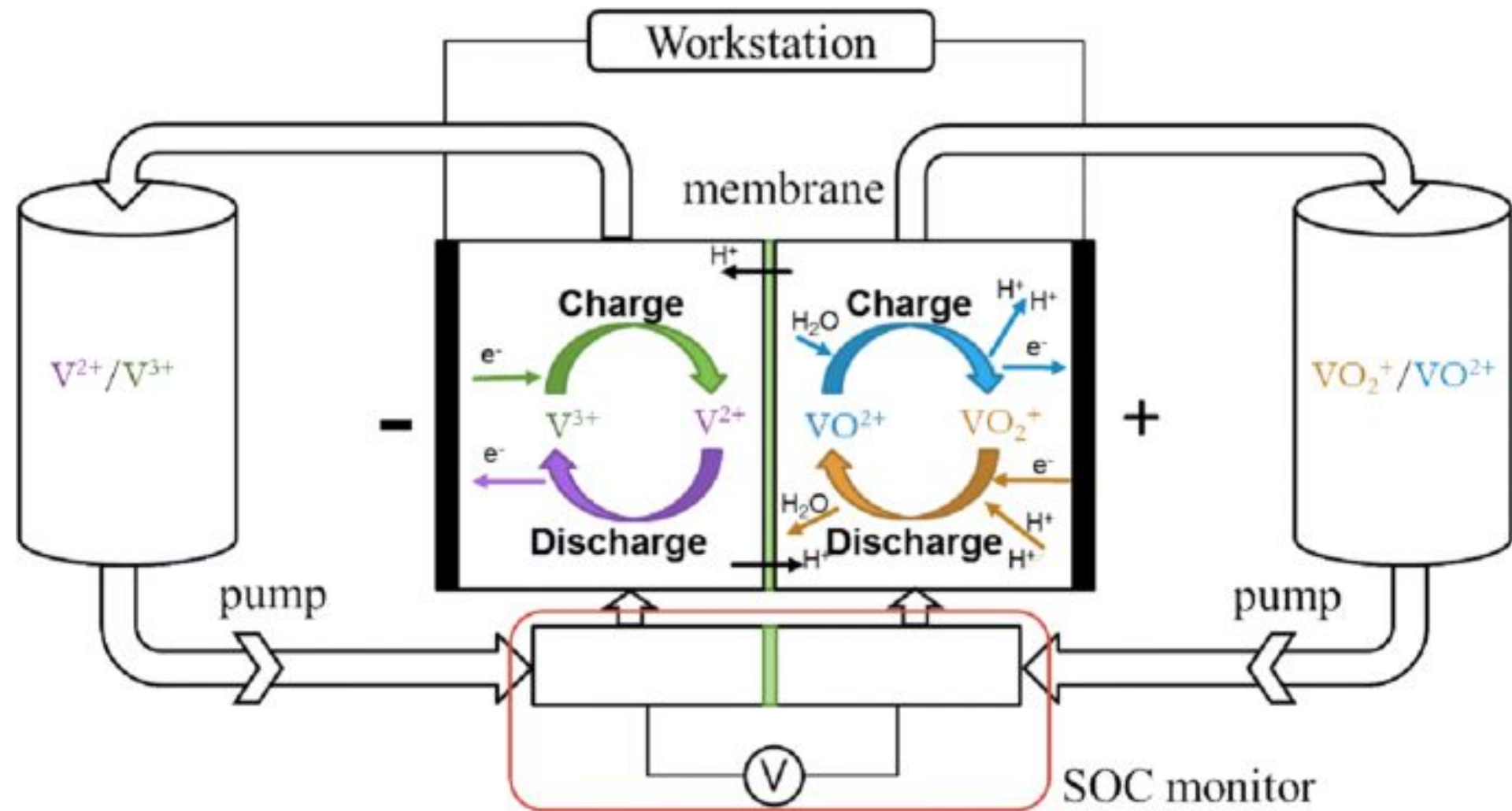
Vanadium flow battery





Vanadium flow battery-animation

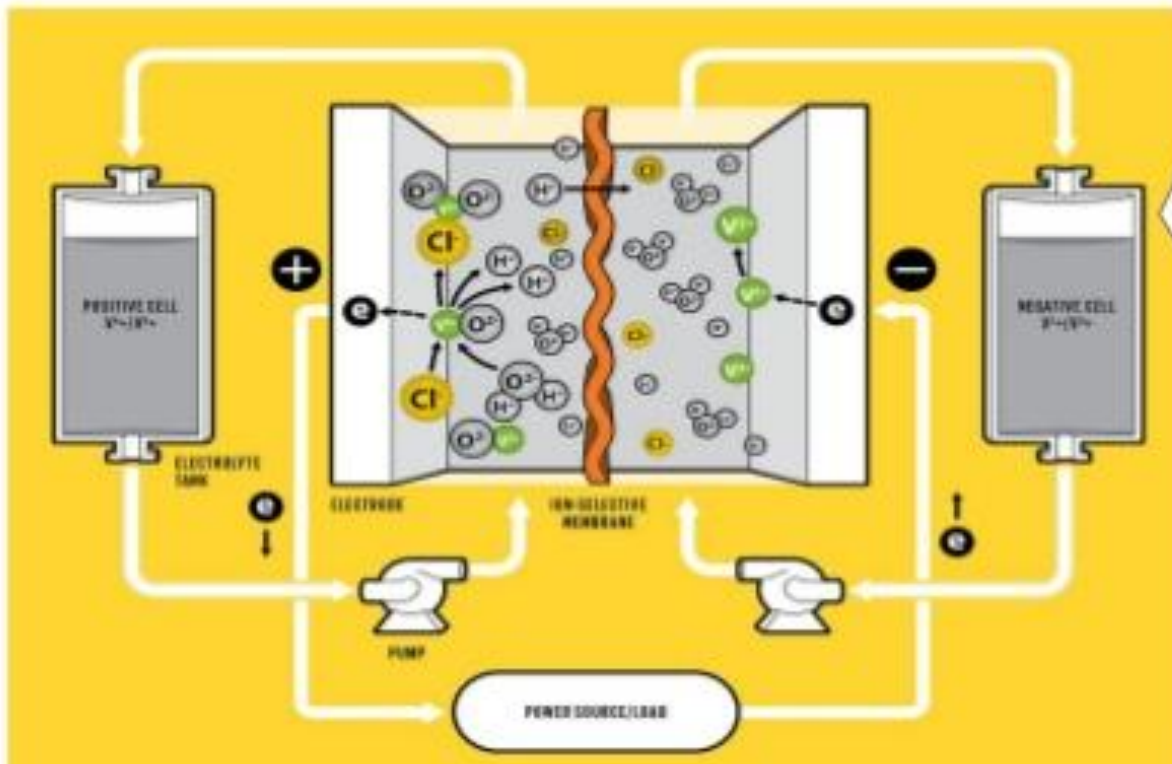
Vanadium flow battery



Working of Vanadium flow battery

How does a vanadium redox flow battery (VRFB) work?

- A flow battery was first developed by NASA in the 1970s and is charged and discharged by a reversible reduction-oxidation reaction between the battery's two liquid vanadium electrolytes
- Unlike conventional batteries, electrolytes are stored in separated storage tanks, not in the power cell of the battery
- During operation these electrolytes are pumped through a stack of power cells, or membrane, where an electrochemical reaction takes place and electricity is produced



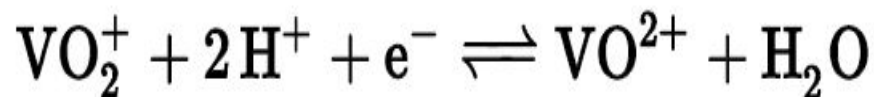
- Vanadium can exist in four different states, allowing for a single element to be used
- Benefits include simplicity and no cross-contamination
- In 2010, US DoE funded research at PNNL yielded an improved electrolyte formula



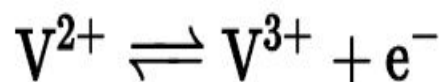


Working of Vanadium flow battery

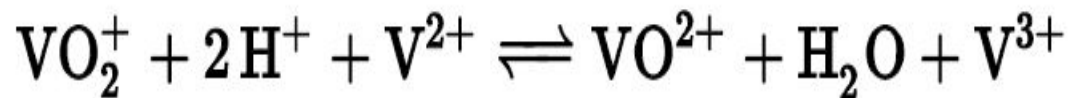
The positive electrode (cathode) half-reaction is



The negative electrode (anode) half-reaction is

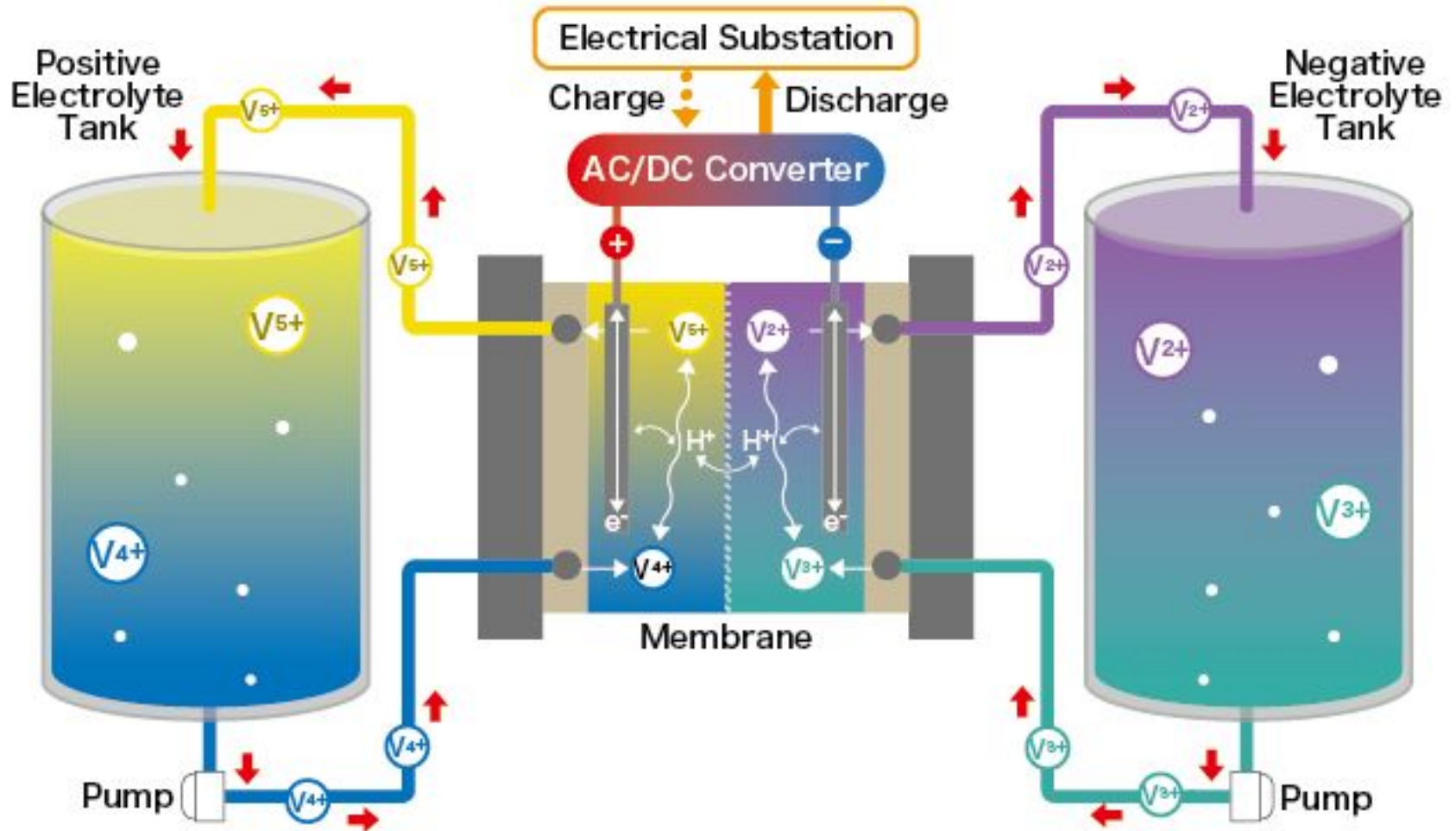


The full reaction (left to right: discharging, right to left: charging) being

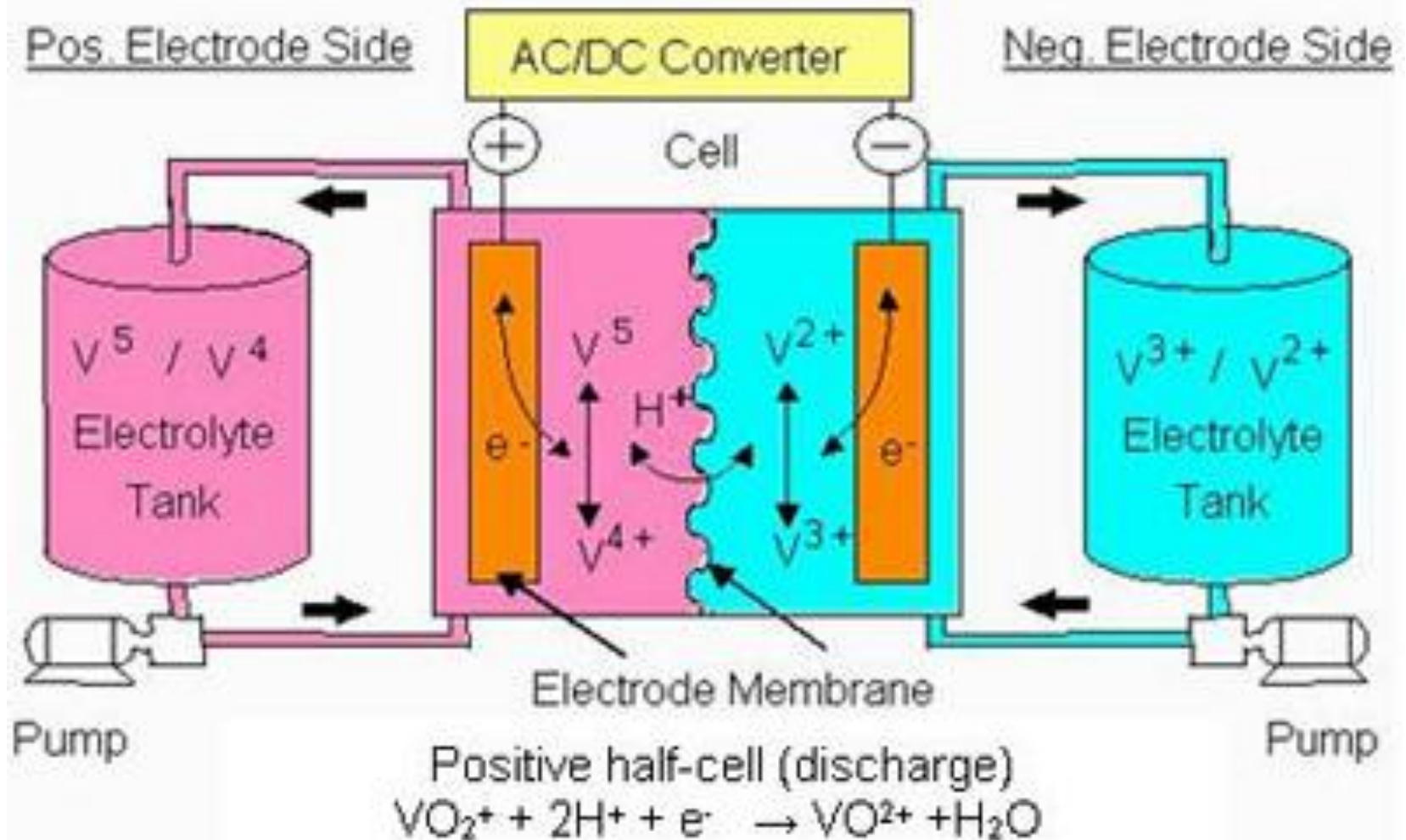


<https://www.youtube.com/watch?v=AagO07cHRG8>

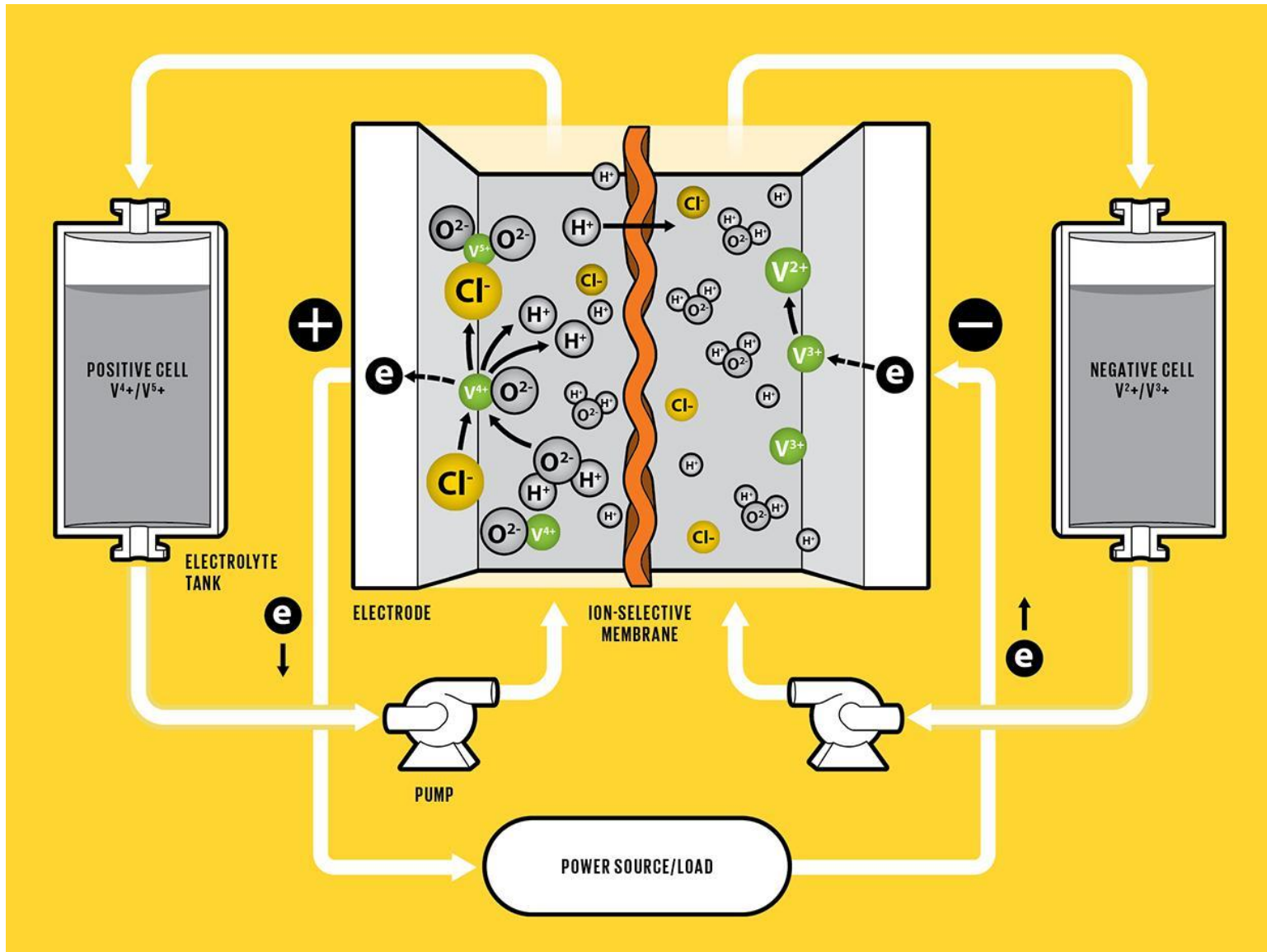
Types of Vanadium flow battery



Vanadium flow battery



Vanadium flow battery





Vanadium flow battery

Long service life: The semi-permanent electrolyte combined with minimal electrode degradation allows for a high number of full charge-discharge cycles before replacement is needed.

No standby loss: During prolonged gaps in use, there is little self-discharge since the charge-carrying electrolyte is stored in separate tanks.

Low maintenance: The charge state of each cell is the same since the same electrolyte is used for all cells, thus overcharging is not necessary to guarantee a uniform a charge.

Recyclability & Safety: Waste vanadium can be reused and cross-contamination across the positive and negative electrode compartments does not affect the composition. Also, the electrolytes are relatively nontoxic.

Charging characteristics: Redox flow batteries are "not affected by fluctuating power demand, repeated total discharge, or charge rates as high as the maximum discharge rates." These actions severely reduce cycle life in other batteries.

<https://www.youtube.com/watch?v=db8nV70hnkU>



Vanadium flow battery: Advantages

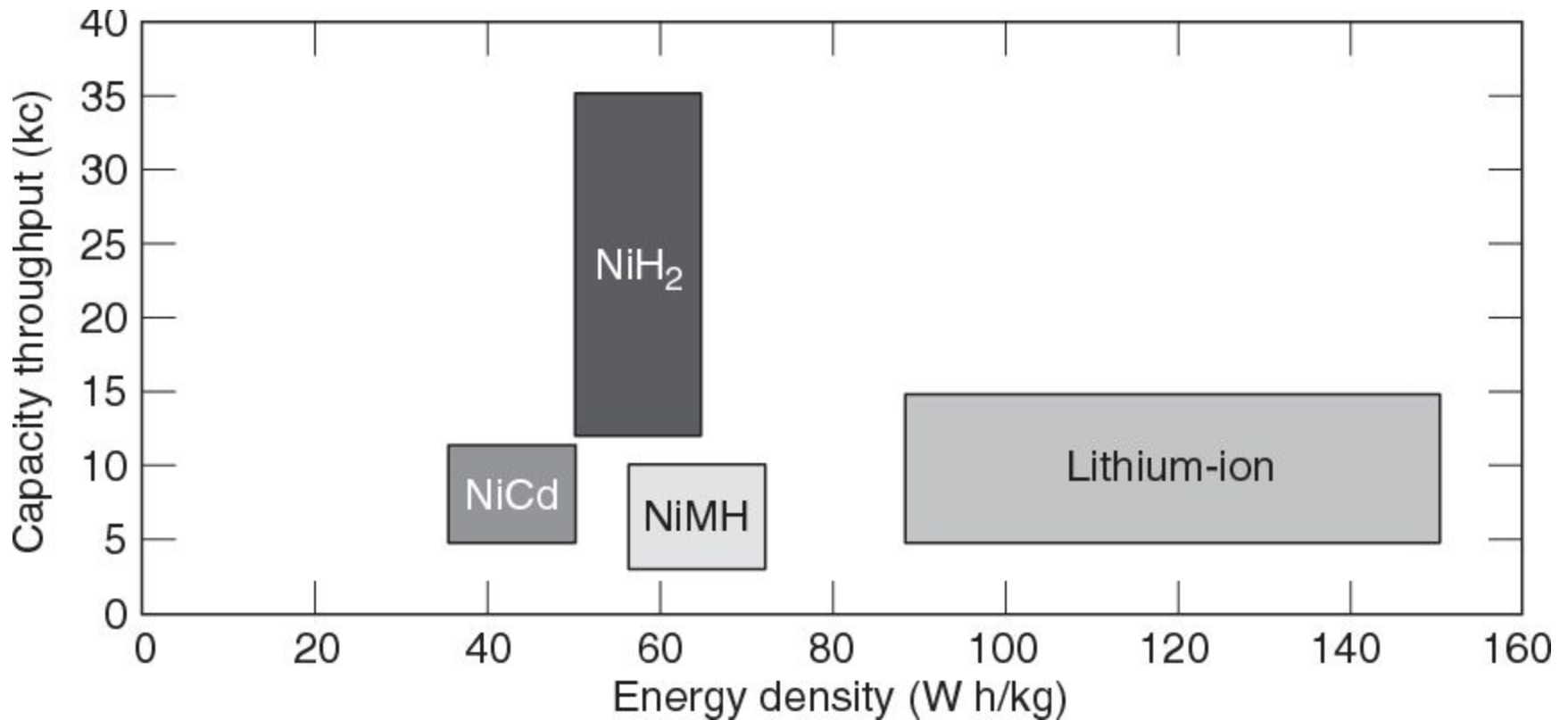
- no limit on energy capacity
- can remain discharged indefinitely without damage
- mixing electrolytes causes no permanent damage
- single charge state across the electrolytes avoids capacity degradation
- safe, non-flammable aqueous electrolyte
- wide operating temperature range including passive cooling
- long charge/discharge cycle lives: 15,000-20,000 cycles.



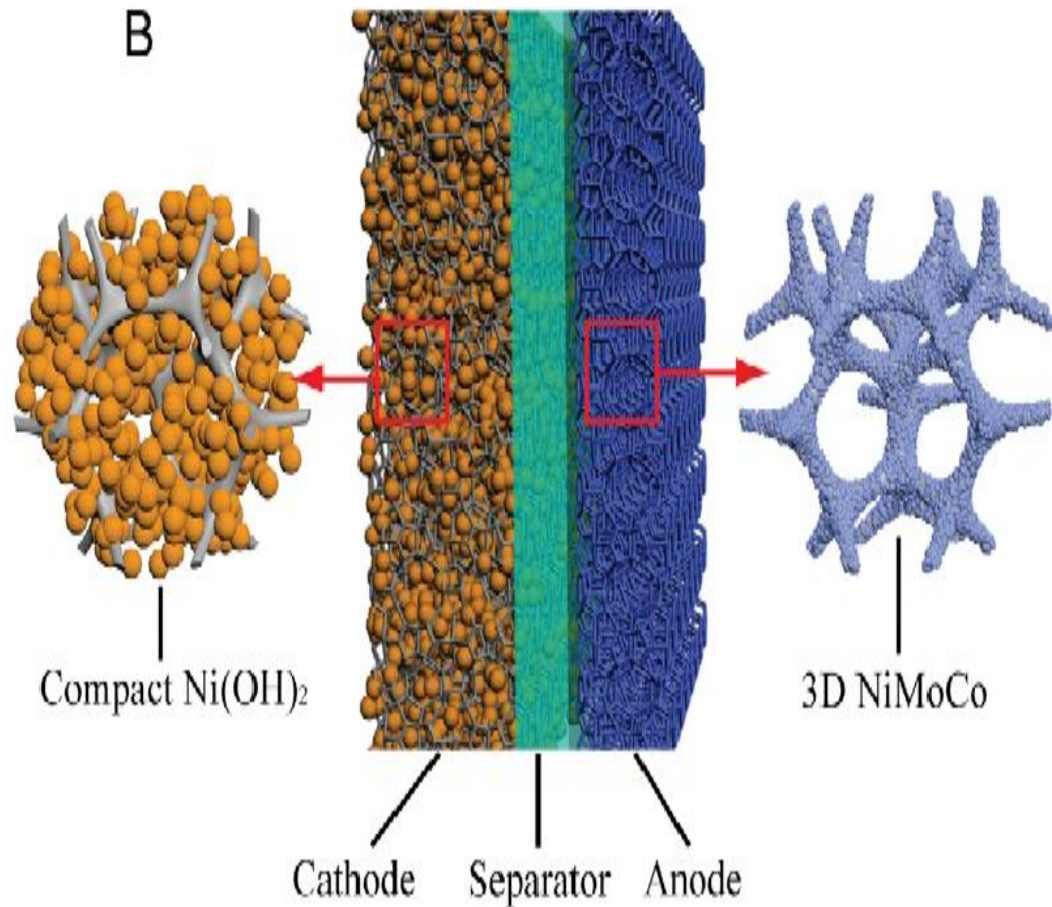
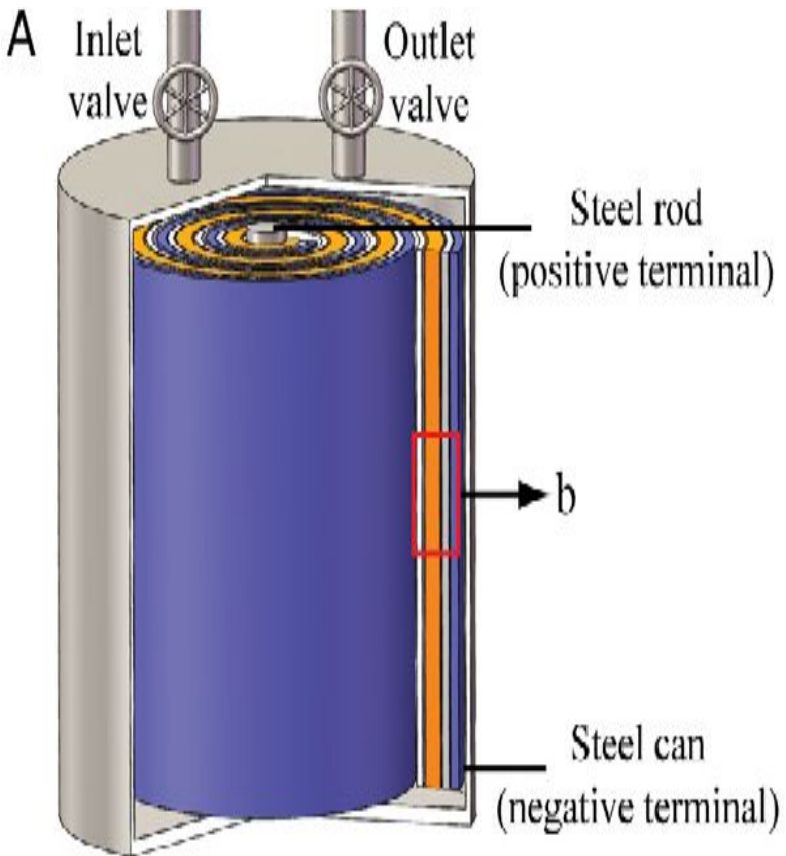
Vanadium flow battery: Limitations

- relatively poor energy-to-volume ratio compared to standard [storage batteries](#)
- relatively poor round trip efficiency
- high weight of aqueous electrolyte
- relatively high toxicity of oxides of vanadium

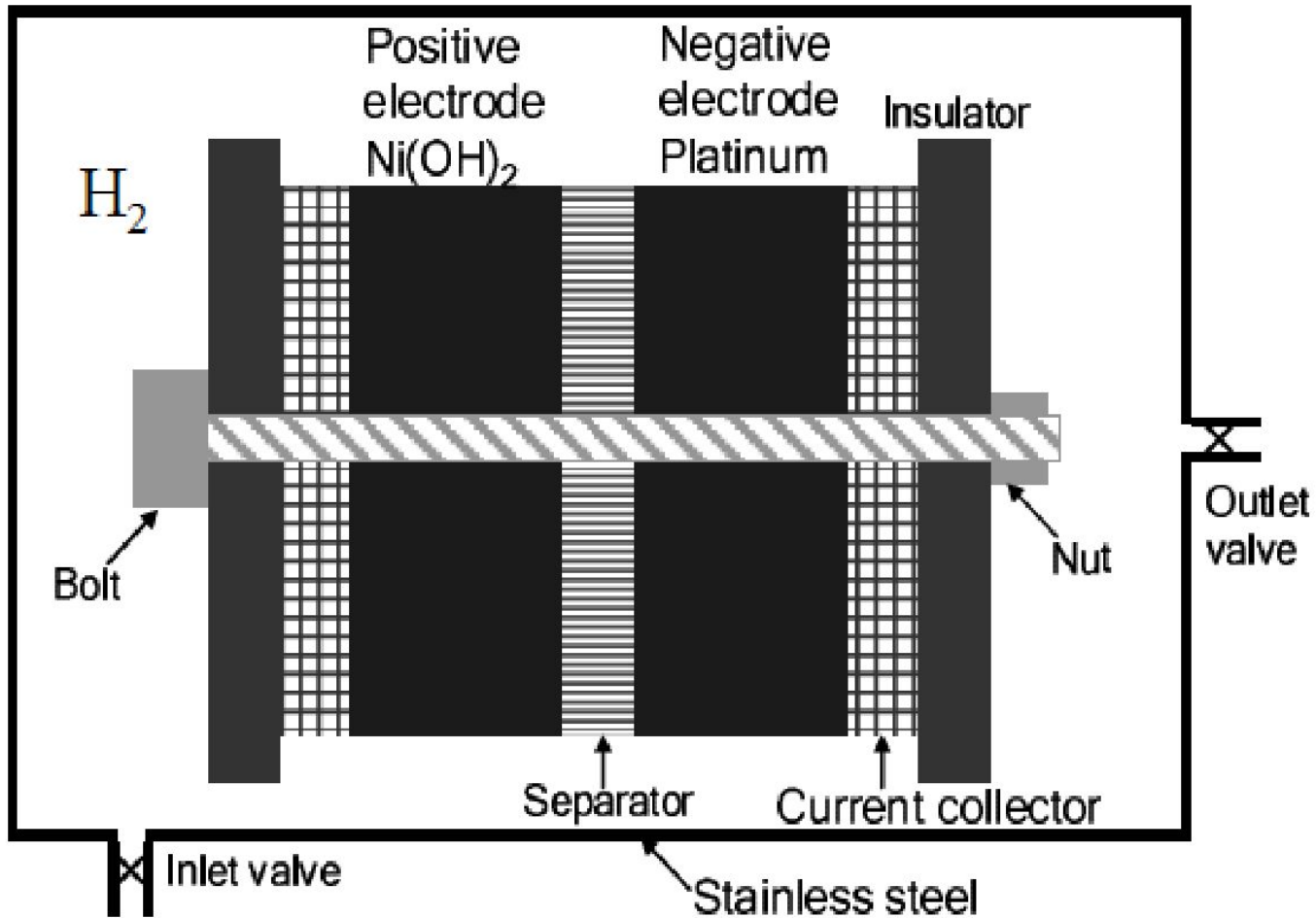
Nickel hydrogen batteries



Nickel hydrogen batteries

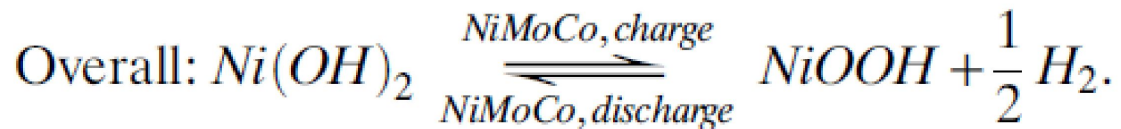
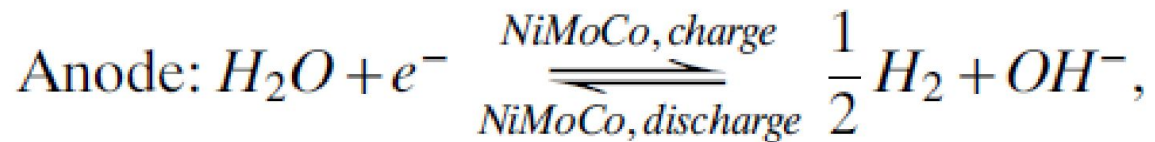
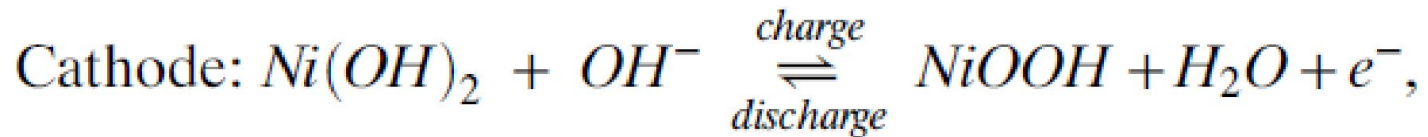


Nickel hydrogen batteries





Nickel hydrogen batteries: reaction



Nickel hydrogen batteries: reaction





Nickel hydrogen batteries

Advantages

Light weight and high gravimetric energy density.

High cycle life (50,000 cycles) and calendar life (15 years)

Can be deep discharged.

Maintenance free.

Because Hydrogen is generated during charging and absorbed during discharging, the gas pressure can be used as a simple measure of the State of Charge (SOC)

Shortcomings

Low volumetric energy density due to gaseous Hydrogen

High self discharge rate

Needs high pressure storage and exotic materials

Limited applications due to high cost.

Applications

Aerospace

Used to power the Hubble Space Telescope